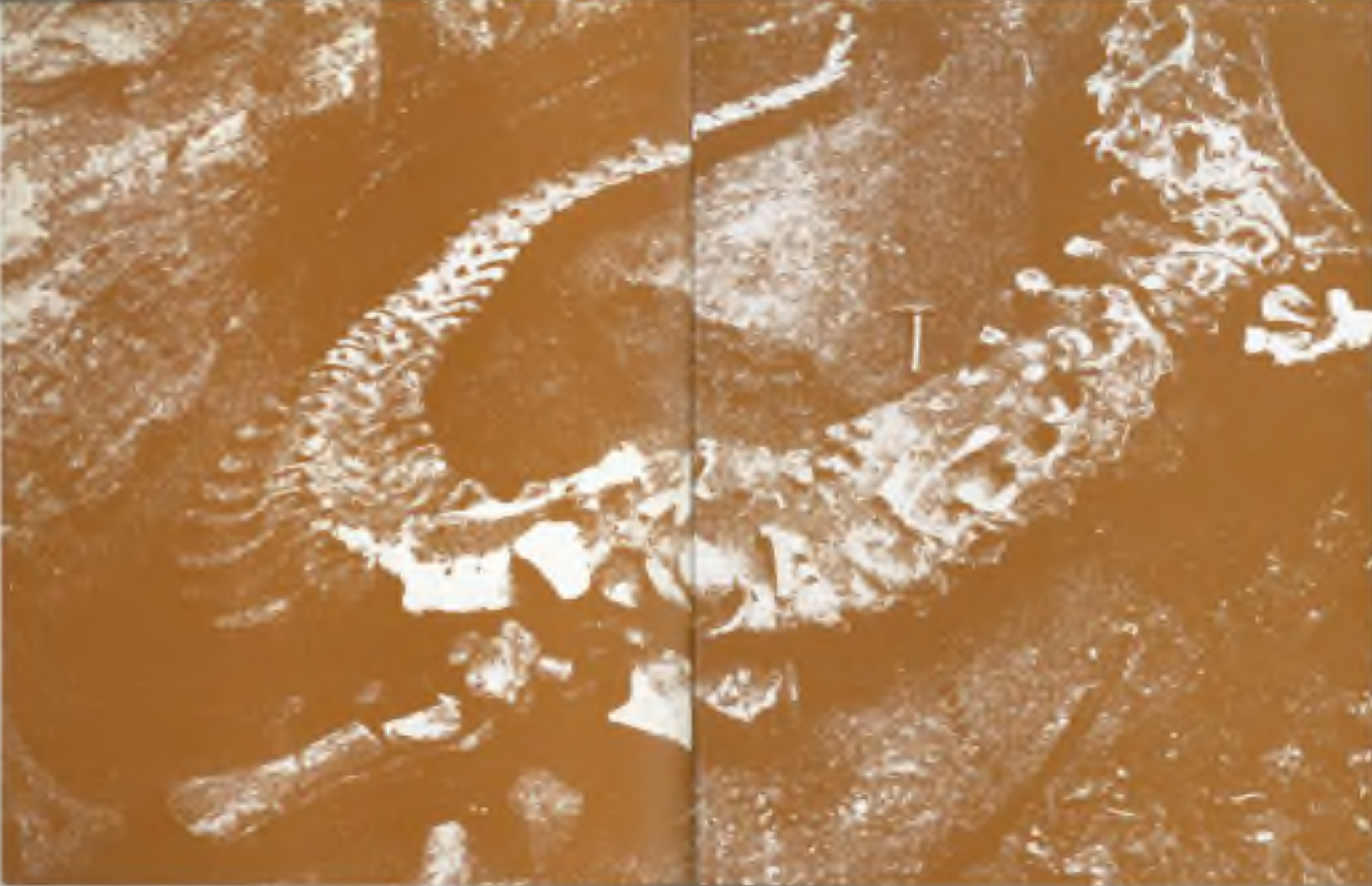


DINOSAURS

From

CHINA





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(Drawn by Peter Trusler).

Inside Front Cover: The skeleton of *Mamenchisaurus hochuanensis*
being excavated in China.

Back Cover: Restoration model of *Rhamphorhynchus*.

D I N O S A U R S

From C H I N A

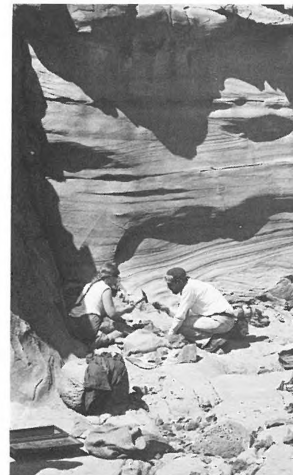
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including
Catalogue to the exhibition
in
The National Museum of Victoria and
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Published by the Council of the National Museum of Victoria
Melbourne — 1982



Excavation by National Museum staff at Dinosaur Cove, Otway National Park, southern Victoria.

Foreword

This book accompanies an exhibition of dinosaurs arranged by the Institute of Vertebrate Paleontology and Paleoanthropology of the People's Republic of China, Beijing; the Australian Museum, Sydney; and the National Museum of Victoria. The exhibition shows two very fine dinosaurs belonging to the Institute and specimens and material from Australia and elsewhere illustrating the nature of dinosaurs and the period in the Earth's history known as the Age of Reptiles.

The three institutions have been working for some time to create this educational and entertaining display in Australia. It was first discussed during informal conversations between Chinese and Australian palaeontologists in 1980 and became a firm prospect early in 1982 when cooperation was formally agreed between the Museums and the Institute's parent body, Academia Sinica in Beijing. Part of the agreement calls for exact copies, or replicas, of the two Chinese dinosaurs to be made in Melbourne and returned to China with the originals at the close of the exhibition. The replicating of these skeletons is expensive and time-consuming, requiring over a year. The method used is illustrated in a section of this book.

The exhibition allows visitors to see for the first time in Australia, the size of the largest land animals that have existed, and to appreciate the diversity of their types. Programmes for schools and other activities are being run in conjunction with the exhibition to further an understanding of the fossil record of the Earth's history and of dinosaurs in particular. For example, in this book you will read of the habitats and the plants and other animals with which the dinosaurs lived. There is a section reviewing that perplexing and still unanswered question: why did the dinosaurs become extinct?

A great deal more is known about dinosaurs than can be told in this book or in the exhibition. How the evidence of the rocks is interpreted to give clues to past climates, and how the age of rocks is determined are fascinating subjects only touched on in this book. For those wishing to follow up some of these matters, or share in the adventures the expeditions hunting for dinosaur remains have faced, a list of recommended books is included with this catalogue.

The Institute and the Museums could not have offered the display without considerable assistance from many individuals and some corporate organisations. In particular we gratefully acknowledge Esso Australia Ltd as official sponsor of the Exhibition. We also wish to acknowledge the Utah Foundation for making it possible to produce the replicas to be retained in Australia, and to the following companies for their generous support: Cathay Pacific Airways Ltd., Dow Corning Australia Pty. Ltd., Kellogg (Aust.) Pty. Ltd., Kodak (Australasia) Pty. Ltd., McDonald's Family Restaurants, Trans Australian Airlines, The Ian Potter Foundation, The Sidney Myer Charitable Trust, the Australia-China Council and Thomas Baker (Kodak), Alice Baker and Eleanor Shaw Benefactions.

Professor J. W. Warren,
President, Council of the
National Museum of Victoria

What are Dinosaurs

Probably no creatures that ever walked the face of the Earth have so stirred human imagination as the DINOSAURS, but many ideas about them are incorrect. Even their name is misleading: it means "terrible lizard" from the Greek "dinos" meaning terrible and "sauros" meaning lizard, but neither of the two groups of reptiles called dinosaurs are lizards.

During the Age of Reptiles which lasted from 200 million years ago to about 65 million years ago the main reptile groups were the "ruling reptiles" or archosaurs (including dinosaurs, pterosaurs and crocodiles) as well as the aquatic plesiosaurs and the dolphin-like ichthyosaurs (Fig. 1).

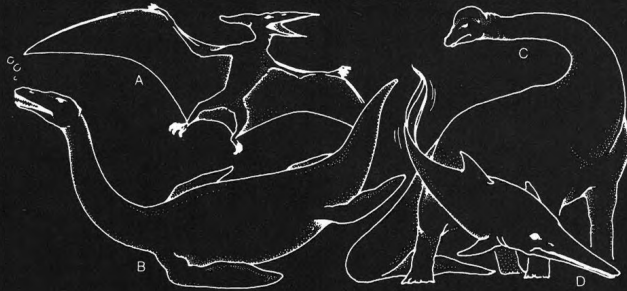


Fig. 1
The principal types of ruling reptiles are (a) pterosaurs, (b) plesiosaurs, (c) dinosaurs, (d) ichthyosaurs.

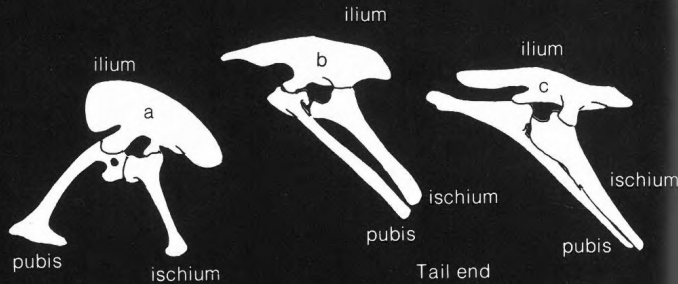


Fig. 2
Hip joints of the two forms of dinosaurs. (a) the lizard-hipped Saurischia, (b) an early bird-hipped Ornithischia and (c) a late bird-hipped Ornithischia, showing the forward projecting part of the pubis.

The dinosaurs can be divided into two groups by the structure of their hip or pelvic girdle (Fig. 2), the saurischians or "lizard-hipped" dinosaurs and the ornithischians or "bird-hipped" dinosaurs. In the Saurischia the hip bones are arranged like those of other reptiles with one of the two bones below the hip joint pointing forwards (the pubis) and the other backwards (the ischium). In the Ornithischia both of these bones point backwards, though in later forms there is a forward projecting part of the pubis. Other differences in the skeletons of the two groups are the presence in the ornithischians of an extra bone at the front of the lower jaw (pre-dentary bone) (Fig. 3) and thin bony rods supporting their back-bone. The saurischians have neither of these characteristics.

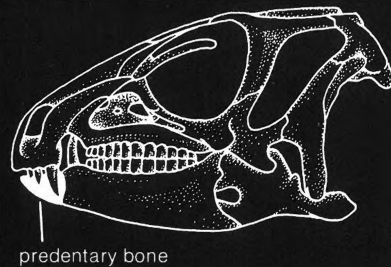


Fig. 3
Skull of an ornithischian showing the pre-dentary bone.

LIZARD-HIPPED (Saurischians)

BIRD-HIPPED (Ornithischians)

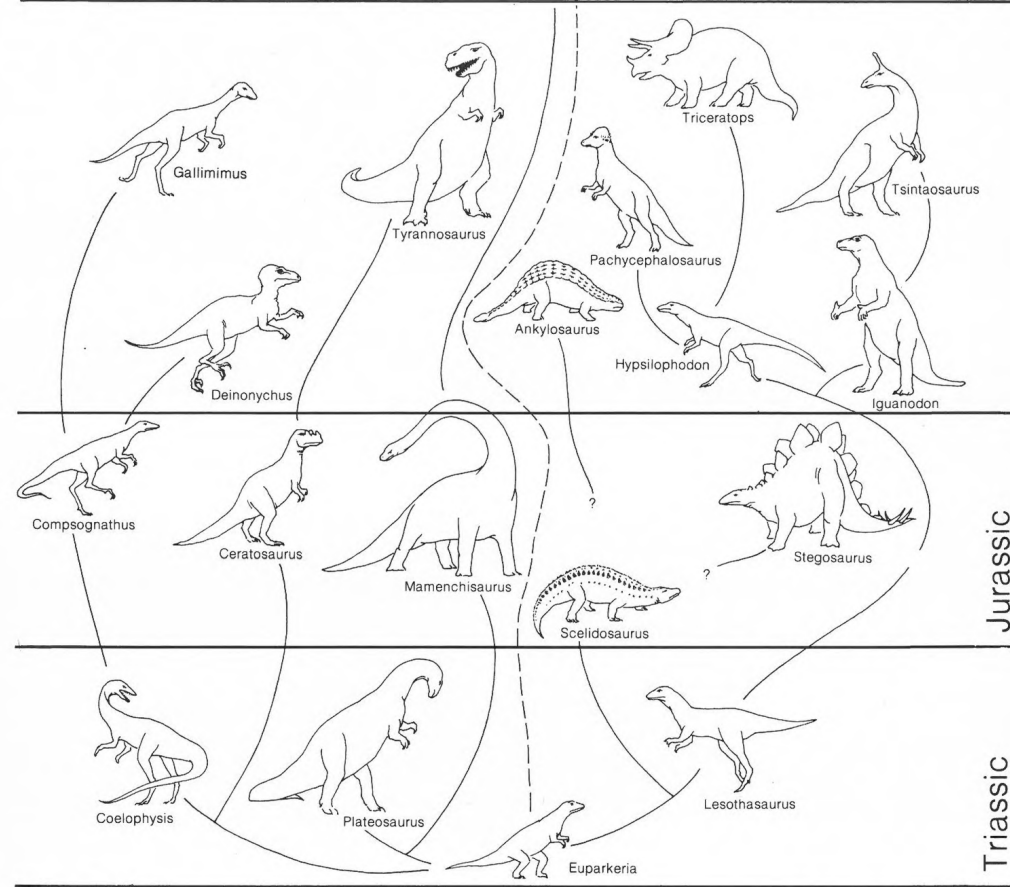


Fig. 4
The dinosaur family tree during the Age of Reptiles.

The Dinosaur Family Tree

The dinosaurs were descendants of a primitive group of archosaur reptiles that lived about 280 to 200 million years ago. These gave rise to the saurischians and ornithischians, each of which diversified into a great variety of forms over the next 130 million years (Fig. 4).

The saurischians included enormous plant-eaters that walked on all fours, sauropods such as *Mamenchisaurus*, and large two legged carnivores, theropods such as *Tyrannosaurus*.

Among the ornithischians were the duck-billed dinosaurs, hadrosaurs such as *Tsintaosaurus*, and the plated dinosaurs, stegosaurs such as *Stegosaurus*. All ornithischians are thought to have been plant eaters.

The Age of Dinosaurs

Contrary to many popular stories, dinosaurs and humans did not live on the Earth at the same time. Our immediate ancestors appeared on the scene less than four million years ago, but dinosaurs lived during the Age of Reptiles (or the Mesozoic Era), which ended 65 million years ago (Fig 5). During that Age, which lasted more than 130 million years the dinosaurs diversified into a great many forms, varying from one the size of a chicken to others that dwarfed elephants. Some walked and ran on their hind legs like humans, while others moved on four feet; many developed heavy armour, probably as a protection against the swift and agile carnivorous dinosaurs (theropods), some of which developed relatively large

brains making them the cleverest of their kind. Other dinosaurs developed a variety of horns and crests. Some may have been warm-blooded, producing their own body heat just as mammals do, although the large, plant-eating, sauropod dinosaurs almost certainly depended on the sun's heat, that is they were cold-blooded. In short, the diversity of dinosaurs may have been every bit as great as that of the mammals today and perhaps even more bizarre.

Continental Drift

When dinosaurs were alive, the Earth was quite different from the one we know.

Fig 5
A chart of geological time showing the main eras.

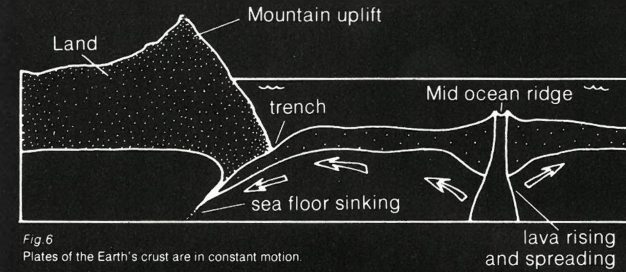
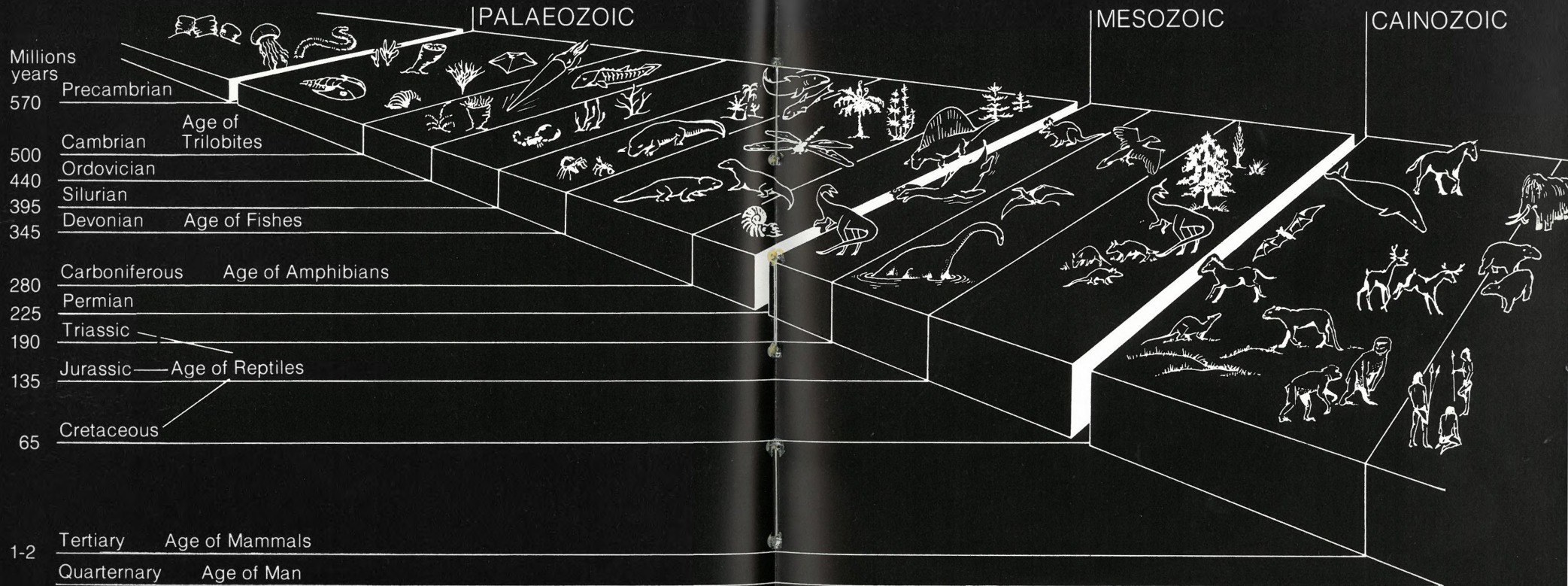


Fig 6
Plates of the Earth's crust are in constant motion.

Continents such as Australia, Antarctica, and Asia probably were not in the positions they occupy today.

The upper kilometres of the Earth's crust appear to be divided into a series of rigid platforms or plates on which ride the continents and ocean basins. Where these huge plates meet one another, new molten material is being added from below (as along oceanic ridges like the Mid-Atlantic Ridge) and the plates are moving apart (Fig 6). In other places (as along oceanic trenches, such as the Marianas Trench in the West Pacific) one plate slips under another and is remelted at great depth. In other places the two plates slip laterally

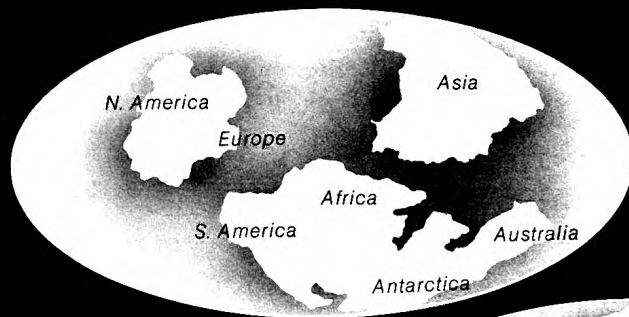
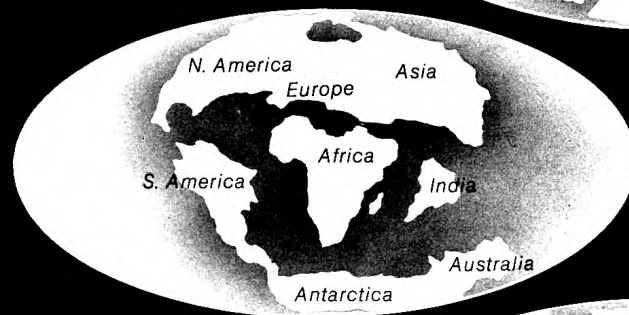
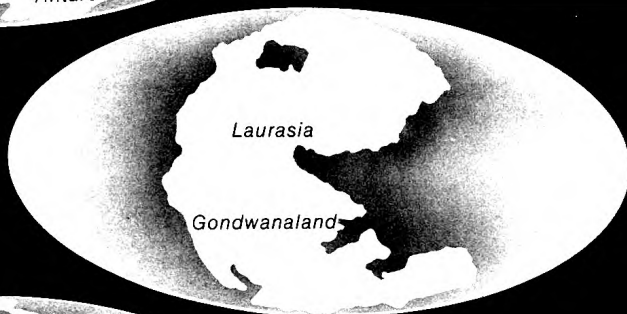


Fig. 7

The process of Continental Drift.

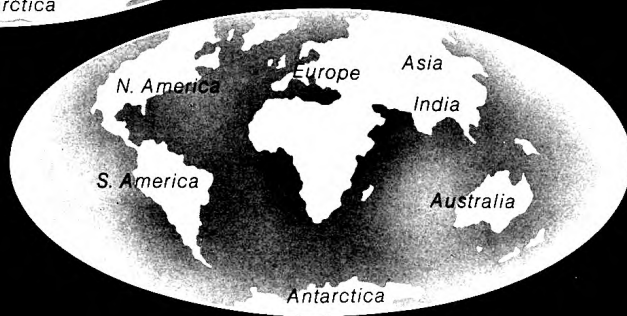
(a) Palaeozoic Era (600 to 225 million years ago). The early arrangement of the plates into three large blocks.

(b) Triassic Period (225 to 195 million years ago). The joining of these blocks into one huge supercontinent, Pangaea.



(c) Cretaceous Period (135 to 64 million years ago). Pangaea first split into two large masses Laurasia and Gondwanaland. These then subdivided into the present continents.

(d) Cainozoic Period (64 million years ago to present). Continents moved into their present positions.



past one another. These kinds of movements have evidently been taking place for at least 700 million years, and probably much longer so that continents have moved relative to one another through geological time. As an example, during the last 50 million years Australia has moved north several degrees of latitude away from Antarctica to its present position nearer to Asia.

At the beginning of the Age of Reptiles, the major land masses we know today were joined into a single continent called Pangaea from the Greek *panthos* meaning "all" and *gaia* meaning "land" (Fig. 7). But by the beginning of the Cretaceous period, 135 million years ago, this super-continent had fragmented: first into Laurasia in the north and Gondwana in the south; then Gondwana further fragmented into smaller pieces, South America, Africa, India, Australia and Antarctica. By 65 million years ago India and New Zealand had split away, but South America, Antarctica and Australia remained tenuously connected to one another until perhaps 50 million years ago. Such continental wanderings controlled the movement of dinosaurs from one part of the world to another and can explain many of the similarities and differences shown by dinosaur faunas in different parts of the world during Mesozoic times.

Climate and Environment

During the time dinosaurs lived, climates differed greatly from those of today. Climates through most of the Mesozoic Era were generally milder than at present and temperatures were higher. Rocks formed in the ocean during the Cretaceous which are now exposed on land far from the sea indicate that the sea level was very high at that time. This suggests there was an absence of glacial ice at the poles because if water is trapped in polar ice it results in a low sea level. Because of the higher sea level many continents were flooded by shallow seas which made the climate milder and affected the geography of the land. For example, because of the distribution of ancient marine rocks such as limestone we know that Australia was a series of islands, not a continuous area of land as we know it today. Only during the later part of the Cretaceous period did climates begin to grow cooler. It was during this cooling phase that the dinosaurs disappeared.

Just as climates were different so were plants. For most of the dinosaurs' reign, no flowers graced the landscape. Flowering plants, called angiosperms, did not develop until sometime in the Cretaceous period, 80 or 90 million years ago. They exploded in variety just at the time of the extinction of the dinosaurs. Before the angiosperms, cycads related to those well known today from many places in Australia, pine-like trees such as *Araucaria* (the Norfolk Island Pine), ferns and tree ferns had dominated the landscape over which the dinosaurs wandered (Fig. 8). Relatives of all these earlier plants still grow in Australia today, but they are overshadowed in number and diversity by the tremendously successful and beautiful plants with flowers.

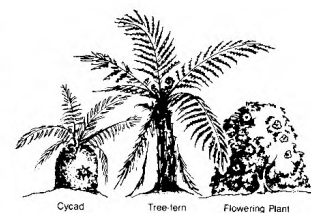


Fig. 8

Two of the land plant types seen during the Age of Reptiles were the cycads and tree-ferns, replaced later by flowering plants.

Why the Dinosaurs Became Extinct

The fossil record shows that dinosaurs and other reptiles that dominated the Earth during the Mesozoic Era became extinct about 65 million years ago. However, throughout the 140 million years that dinosaurs existed, single species and even entire groups of species became extinct from time to time, while new ones appeared and increased in numbers. For example, the stegosaurs, first recorded in the middle of the Jurassic period 160 million years ago were extinct by the end of that period some 25 million years later. But whereas several groups of dinosaurs died out long before the end of the Mesozoic Era, extinction was apparently total at the end of that era even though certain groups such as the duck-billed dinosaurs had been increasing in diversity almost up to the time of their disappearance. Extinction of the last remaining groups of dinosaurs may have occurred over tens of millions of years or may have been more rapid. It may have occurred in no more than a few hundred years, or perhaps in some cases even more suddenly. There is no general agreement on just how quickly the final extinction did occur, but it is an essential point in trying to resolve the causes of the extinction.

Not only did dinosaurs die out, but simultaneously so did a great variety of other living things. These included the winged pterosaurs, the marine ichthyosaurs and plesiosaurs, together with other forms of marine life such as the ammonites and many microscopic single-celled organisms. The world's flora was also radically changed by the disappearance of many non-flowering plants, such as primitive cone-bearing trees and some groups of cycads. Flowering plants, however, continued to flourish and increase in both numbers and diversity.

What global explanation can be offered to account for such widespread extinction? Several theories have been put forward by scientists — from palaeontologists to physicists — but so far there is no one theory that is generally agreed upon. There

are three possible causes for which there is some scientific evidence: (1) a catastrophic event, (2) climatic change and (3) the appearance of mammals as competitors. These three are not necessarily unrelated and perhaps no single cause, but a combination of several, was responsible for the extinctions.

Cosmic Catastrophe

One proposal, that until recently has not received much support from scientists, is the suggestion that the mass extinction was caused by a collision of the Earth with a comet or an asteroid (Fig.9). Now, however,

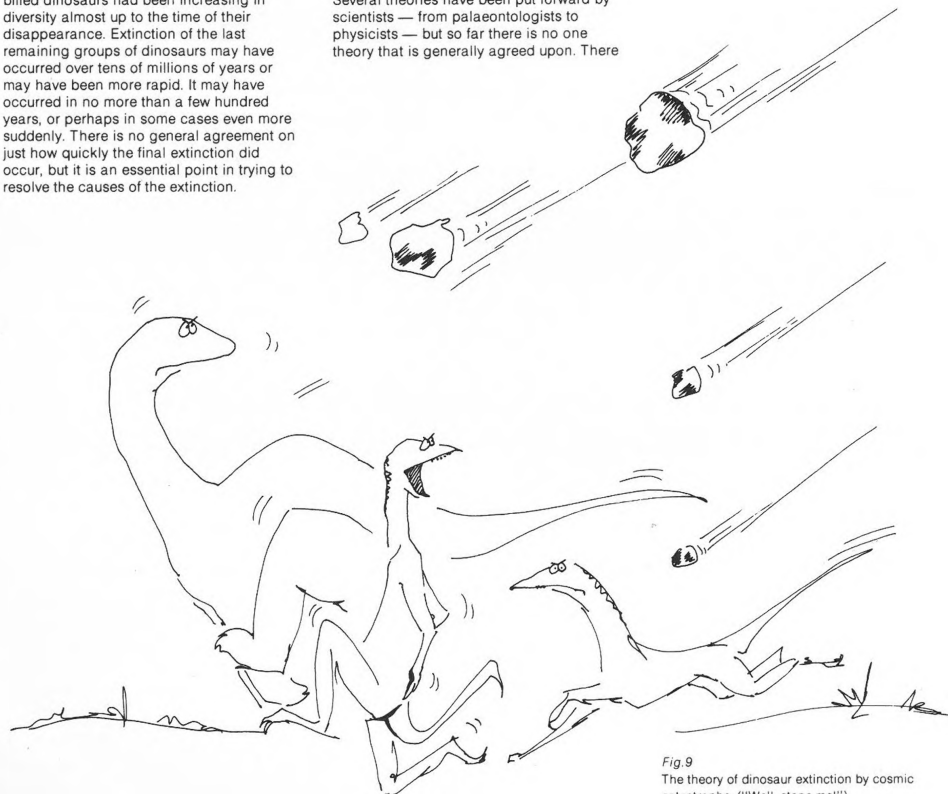


Fig.9
The theory of dinosaur extinction by cosmic catastrophe. ("Well, stone me!")

there is growing evidence of this. It comes from the discovery of concentrations of rare metals, such as iridium, in bands of clay sediments deposited at the end of the Cretaceous period. It is generally thought that the source of these rare metals is extraterrestrial and a collision of the Earth with an asteroid would account for the iridium concentrations appearing in the sediments of one time zone, rather than being spread over many time zones.

Such a collision could have caused extinctions by initially increasing global temperatures at the moment of impact. Following this, dust rich in the rare metals could have filled the atmosphere, blocking out sunlight long enough to suppress photosynthesis and plant growth. Plants may have survived relatively unscathed as seeds, but animals dependent on plants for food would have starved and in turn so would the meat eating predators. Furthermore, the presence of dust of high concentrations of rare metals such as iridium would have acted as biological poisons, perhaps leading to further extinctions.

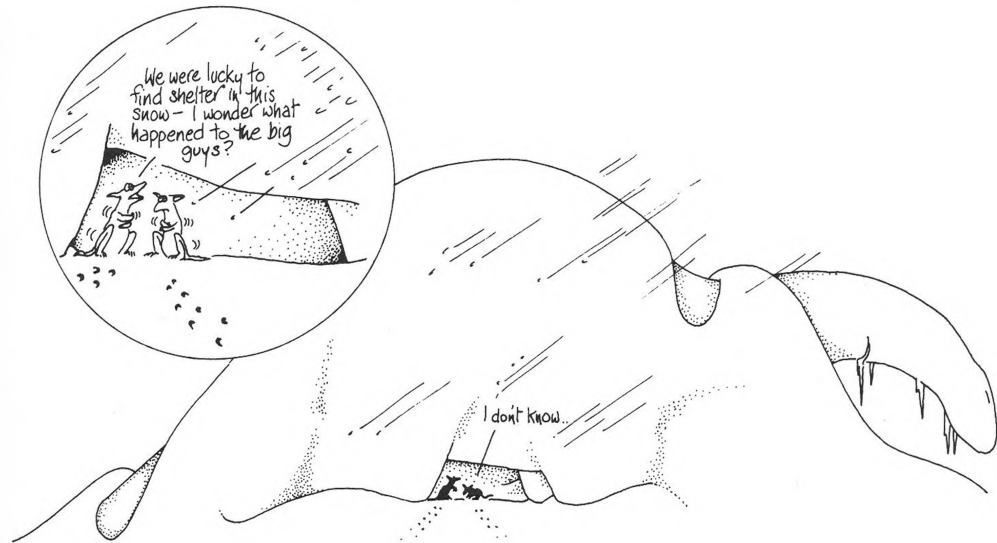


Fig.10
The theory of dinosaur extinction by climatic change. (Melbourne weather again.)

This theory would account for the general demise of many types of organisms, both large and small, land living and aquatic, but requires the extinctions to be rather sudden and this cannot yet be shown to be the case, for all those groups that failed to survive beyond the end of the Cretaceous.

Climatic Change

Several long-standing theories are based on suggestions of gradual climatic change, either warming or cooling. Marked temperature changes may leave clear evidence in the geological record, for example the scars where ancient glaciers moved indicate ice-age conditions, but moderate temperature changes of the order suggested for the close of the Cretaceous are difficult to detect, the evidence being only circumstantial.

Nevertheless, the geological record shows that shallow seas flooding the low-lying continents during the Cretaceous period began to recede, that more land was being exposed and new mountains formed. Widespread, shallow seas have a moderating effect on climate, so as the continents were

drained harsher climates and cooler conditions gradually prevailed. (Fig.10). This must have been detrimental to reptiles, including dinosaurs, that require warm conditions to maintain bodily functions.

Also the draining of the continental seas decreased the extent of swampy lowlands on which many of the plant eating dinosaurs depended. And again, any decrease in plant eaters affects those predators that depend upon their flesh for food.

The Appearance of Mammals

Mammals differ from reptiles in several ways. One important difference is their ability to produce sufficient heat internally to maintain the high level of body temperature necessary to sustain active life. Reptiles also require a high body temperature for normal functioning, but cannot produce sufficient heat for this internally and, therefore, depend on warm atmospheric conditions and radiation from the sun. For this reason mammals are sometimes referred to as being "warm-blooded" and reptiles as "cold-blooded", though in truth reptiles require a body temperature as high, and in some cases even higher, than mammals. This is why any cooling of the climate would detrimentally affect reptiles, but it would not have the same effect on mammals.

Mammals first appeared on earth about 180 million years ago as small, cat-sized animals and they lived in the company of dinosaurs for the remainder of the Age of Reptiles. If conditions were cooling at the end of the Cretaceous period, with climatic belts becoming narrower forcing the "cold-blooded" dinosaurs more into tropical regions, this would have benefitted the "warm-blooded" mammals that could be active in colder climates and even at night. The more active mammals with their numbers and diversity of types on the increase could exploit limited food resources more efficiently than the dinosaurs. In effect, the dinosaurs were losing the struggle for existence to the mammals who survived and thrived to become one of the dominant groups on Earth today.



Dinosaurs in Australia

In the world-wide search for dinosaur remains, Australia is one of the last frontiers. Australian finds have been extremely few, fragmentary, and widely scattered; no Australian deposits of dinosaurs yet discovered rivals, either in quality or quantity of material, those in North and South America, Eurasia and Africa. Only Antarctica, where no dinosaurs have yet been discovered, has fewer than Australia. Despite this, sufficient material is now available to demonstrate that dinosaurs were present in Australia in Triassic, Jurassic and Cretaceous times (200-65 million years ago).

Apart from recent Victorian discoveries most Australian finds have been in Queensland. New South Wales and South Australia have produced only a few isolated bones. All of the specimens come from freshwater or shallow marine sediments, such as those of the early Cretaceous opal-bearing deposits of New South Wales and South Australia. A single, small dinosaur toe-bone has come from White Cliffs, N.S.W. Lightning Ridge, N.S.W., has yielded several opalized vertebrae, teeth and other bones of *Walgettosuchus woodwardi* and *Rapator ornitholestoides*, carnivorous theropods, and *Fulgurotherium australe*, a small herbivore.

In Queensland the skeletal remains of a variety of different dinosaurs have come to light, and the potential for more discoveries is considerable. Several sauropod dinosaurs have been found. *Rhoetosaurus brownei* from Jurassic rocks near Roma is one of the oldest, represented by a considerable part of a skeleton. *Austrosaurus mckillopi*, another sauropod, comes from Maxwellton, central Queensland from early Cretaceous rocks. The most complete skeleton discovered so far is that of an iguanodont-like ornithopod, about 7 m. long, *Muttaburrasaurus langdoni* from the Thomson River area, central Queensland. A partial skeleton of a small, armoured dinosaur, *Minmi paravertebra*, is thought to be an ankylosaur, the first such discovery in Australia and the most complete example known from the southern hemisphere.

Early Cretaceous rocks in Victoria, in the Strzelecki and Otway Ranges near Melbourne, have produced *Allosaurus* (a theropod), an ankylosaur, and two other smaller herbivores similar to *Hypsilophodon*. Coastal rock platforms, which are constantly being eroded by wave action, are the best sites for finding dinosaurs in Victoria. Near Lion Head, west of Cape Otway, a single small theropod dinosaur footprint has been found, but so far in Australia, no dinosaur eggshell has been collected, often an abundant fossil elsewhere in the world.

Near Winton in central Queensland there is evidence that some types of dinosaurs were present in large numbers. A slab of rock has spectacularly preserved the trackways of over 130 individual dinosaurs which are represented by more than 3,300 separate footprints. These dinosaurs were small to medium sized bipedal forms and they were all fleeing in the same direction, apparently in a stampede. Also present on the same slab of rock are the tracks of a large predaceous dinosaur, and perhaps it was its appearance that caused the panic and stampede among the smaller animals.

No doubt there were many other types of dinosaurs in Australia, and the search for these yet unknown forms makes the science of vertebrate palaeontology exciting today, worthy of considerable effort, and one of the rare pursuits of modern science in which both amateur and professional alike can share.

How Fossils are Formed

A fossil is the remains of an animal or plant preserved in rocks. Many species of animals or plants now extinct are known only from the fossil record.

In order for an organism to be found as a fossil, its remains must have gone through a series of stages (Fig. 11).

- After death the organism must be protected from decay or destruction by scavengers. Often it is buried rapidly in the soft silts or muds in a swamp or lagoon.
- The sediment in which the organism is preserved must remain undisturbed and be gradually transformed over a long period of time into rock.
- During the process of rock formation all or part of the bone in the skeleton may be added to or even replaced by minerals in the process known as petrification. Usually only the hard parts of an organism, such as bone, teeth or shell are preserved as fossils but when conditions are at their most favourable other structures may be preserved, eg. impressions of skin and internal organs.
- Rocks containing fossils are subject to all the movements in the Earth's crust over geological time, which include the uplift, folding and buckling involved in the processes of mountain building. To be found easily fossils have to be located where they are likely to be uncovered by some natural or accidental process.

Fossils may be uncovered and so brought to the attention of the collector by weathering. Rain, wind, and sun cause rocks to weather away and expose the fossil. If not discovered at this stage, it too is weathered away and lost.

Fossils may be uncovered by quarrying or by excavation for building or construction or by scientific "digs" for fossils.

To be studied and preserved the fossil must come to the attention of a palaeontologist.

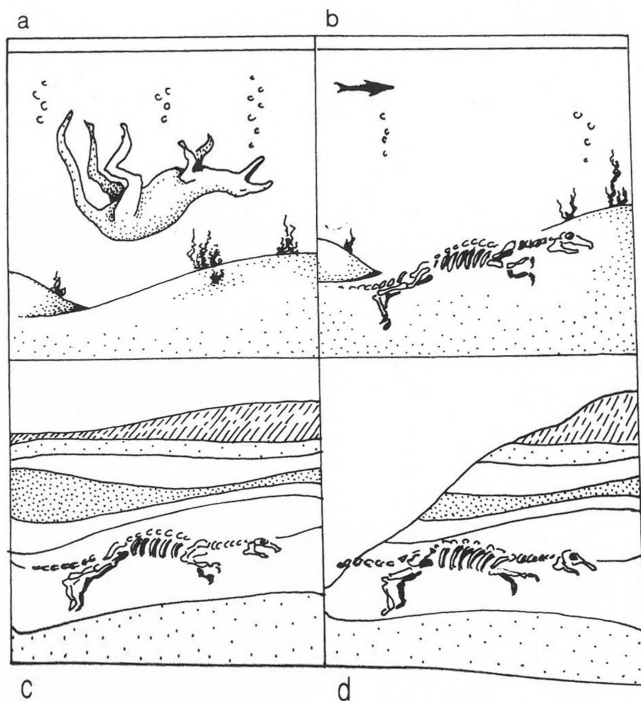


Fig. 11

The stages necessary for fossil formation; (a) the animal falls into a place where it is preserved; (b) sediment covers the remains; (c) the remains are incorporated into the rocks; (d) the rocks are weathered or disturbed exposing the fossil.

Catalogue

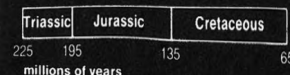
This section is a catalogue of the 'Dinosaurs from China Exhibition'.

Though the exhibition contains specimens of all the major groups of dinosaurs together with examples of pterosaurs (flying reptiles) and a primitive bird, many from the collections of the National Museum of Victoria and the Australian Museum, the "star specimens" are the two large skeletons of dinosaurs from China.

Each specimen in the exhibition is described. Its scientific name and its meaning and pronunciation are given. The geological period in which the species lived and any special features of the specimen and the species are set out.

The age of the specimen is indicated by a shaded area on the time scale marked in millions of years before the present (Fig. 12).

Fig. 12 (m.yrs. — millions of years)



Contents of the Exhibition

Listed below are the specimens in the exhibition with the page number in this book on which further information about the species can be found.

36	Allosaurus —
37	Ankylosauria —
46	Archaeopteryx —
38	Camptosaurus —
41	Coelophysis —
43	Deinocoelurus —
44	Dzungaripterus —
22	Gallimimus —
33	Homalocephale —
39	Hypsilophodon —
18	Mamenchisaurus —
45	Psittacosaurus —
31	Pteranodon —
31	Rhamphorhynchus —
35	Rhoetosaurus —
32	Saurolophus —
40	Stegosaurus —
34	Supersaurus —
20	Tsintaosaurus —
42	Tyrannosaurus —

The Chinese Dinosaurs

Mamenchisaurus hochuanensis (ma-men-chi-sawr-us)

Meaning: Mamenchisaurus is the locality in the People's Republic of China where this kind of dinosaur was first found.

Age: Late Jurassic (140-155 m.yrs.)

Diet: Plant-eater

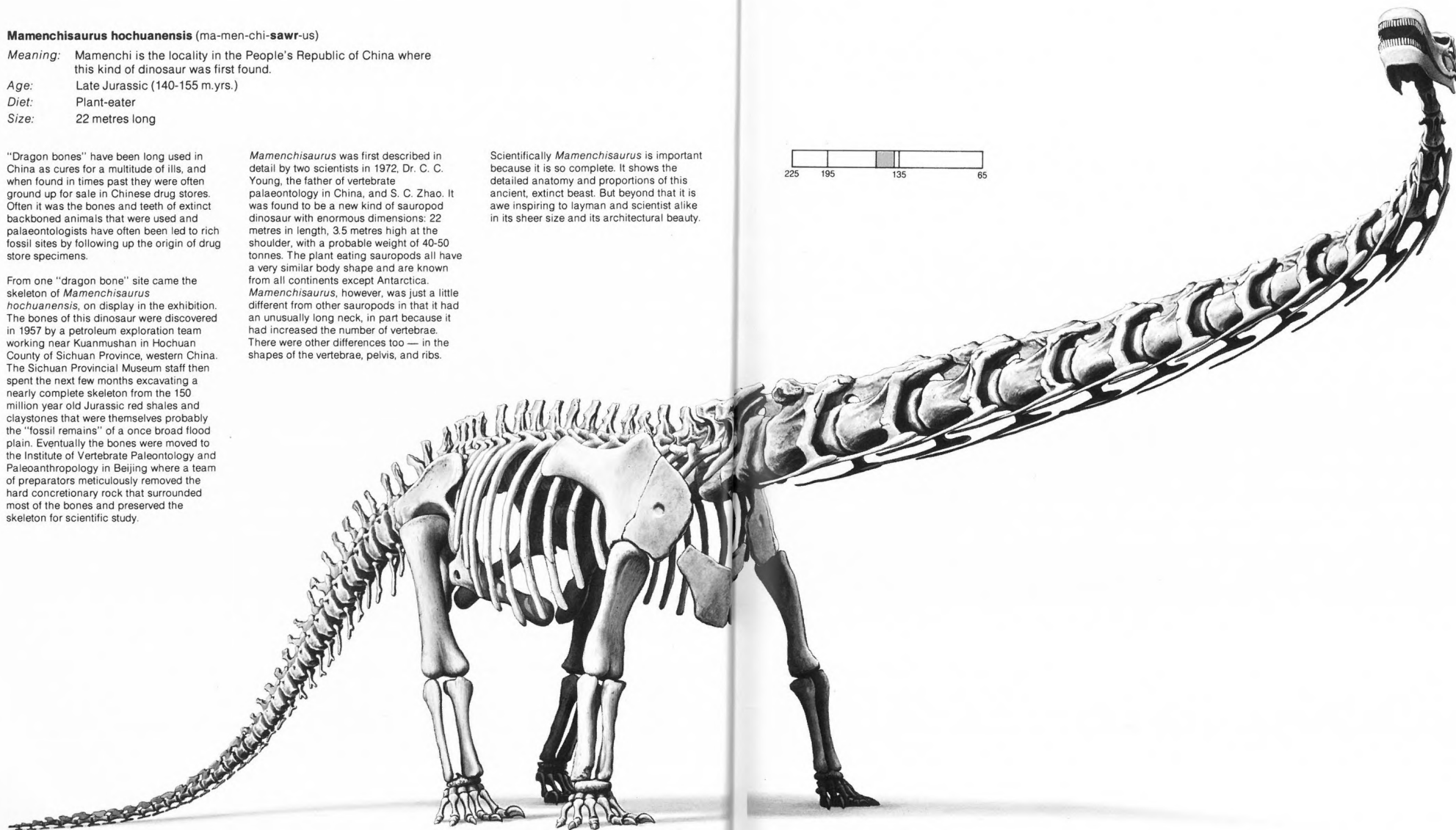
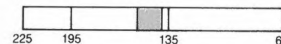
Size: 22 metres long

"Dragon bones" have been long used in China as cures for a multitude of ills, and when found in times past they were often ground up for sale in Chinese drug stores. Often it was the bones and teeth of extinct backboned animals that were used and palaeontologists have often been led to rich fossil sites by following up the origin of drug store specimens.

From one "dragon bone" site came the skeleton of *Mamenchisaurus hochuanensis*, on display in the exhibition. The bones of this dinosaur were discovered in 1957 by a petroleum exploration team working near Kuanmushan in Hochuan County of Sichuan Province, western China. The Sichuan Provincial Museum staff then spent the next few months excavating a nearly complete skeleton from the 150 million year old Jurassic red shales and claystones that were themselves probably the "fossil remains" of a once broad flood plain. Eventually the bones were moved to the Institute of Vertebrate Paleontology and Paleoanthropology in Beijing where a team of preparators meticulously removed the hard concretionary rock that surrounded most of the bones and preserved the skeleton for scientific study.

Mamenchisaurus was first described in detail by two scientists in 1972, Dr. C. C. Young, the father of vertebrate palaeontology in China, and S. C. Zhao. It was found to be a new kind of sauropod dinosaur with enormous dimensions: 22 metres in length, 3.5 metres high at the shoulder, with a probable weight of 40-50 tonnes. The plant eating sauropods all have a very similar body shape and are known from all continents except Antarctica. *Mamenchisaurus*, however, was just a little different from other sauropods in that it had an unusually long neck, in part because it had increased the number of vertebrae. There were other differences too — in the shapes of the vertebrae, pelvis, and ribs.

Scientifically *Mamenchisaurus* is important because it is so complete. It shows the detailed anatomy and proportions of this ancient, extinct beast. But beyond that it is awe inspiring to layman and scientist alike in its sheer size and its architectural beauty.



Tsintaosaurus spinorhinus (chin-ta-sawr-us)

Meaning: Reptile from Tsintao
Age: Late Cretaceous (65-95 m. yrs.)
Diet: Plant-eater
Size: 7 metres long

Tsintaosaurus spinorhinus was collected from Late Cretaceous rocks in Laiyang County, Shandong Province, southeast of Beijing. It was a medium-sized dinosaur that was 7 metres in length, 5.5 metres in height, and probably weighed about 3 tonnes in life.

Like *Mamenchisaurus* it was a plant eater, but instead of having small peg-like teeth, it had a massive battery of grinding teeth that would have efficiently dealt with tough foliage, even that containing the abrasive material silica. *Mamenchisaurus* was probably restricted to a diet of quite soft plant material, whereas *Tsintaosaurus* could have easily dealt with fodder as tough as pine needles or cycad fronds.

Tsintaosaurus was an ornithischian or bird-hipped dinosaur that often stood upright on two hind legs. It belongs in the hadrosaur or duck-billed dinosaur group, which flourished near the end of the Age of Reptiles, during the Cretaceous Period. Relatives are known in North America and eastern Asia, where they were highly diverse, and in Europe, where they are poorly known. There are no known relatives in the Southern Hemisphere.

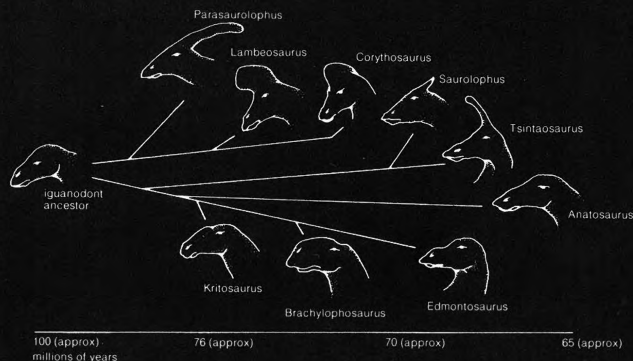
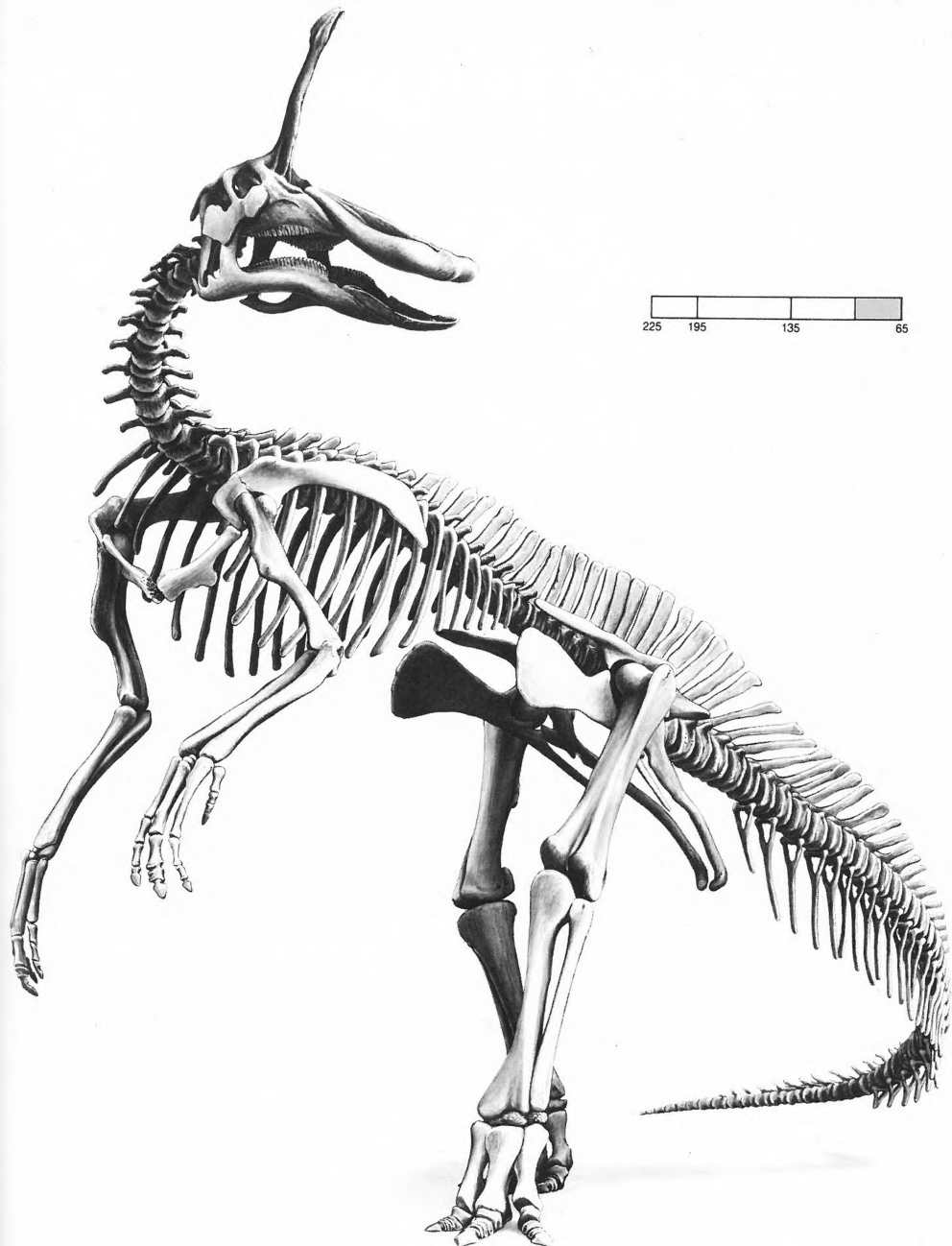
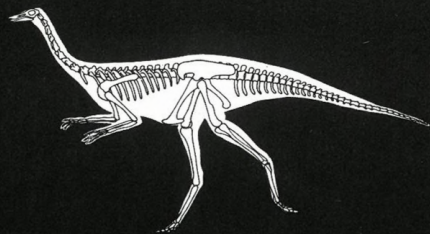


Fig. 13
 A family tree of the duck-billed dinosaurs showing development and possible relationships within the group.

The group is characterized by a whole range of horns, crests, and ridges that are developed on the skulls of different species and, in fact, *Tsintaosaurus* derives its specific name *T. spinorhinus* from the fact that it has a well-developed, solid horn (an extension of the nasal bone) above the skull. This is similar to *Saurolophus*, a close relative and possibly supported a crest to enable the members of the species to recognise each other.

Duck-billed dinosaurs are unknown in Australia, but a group related to them, the iguanodonts, are known from central Queensland. *Muttaburrasaurus*, an iguanodont, is known from a nearly complete skeleton lacking only parts of the tail. It had an expanded hollow chamber on its snout that might have been used to produce mating calls or perhaps to enhance the sense of smell.





Gallimimus (gal-im-ee-mus)

Meaning: 'Chicken-mimicking'; so named because the neck bones have similarities to chicken-like birds.

Age: Late Cretaceous (65-95 m. yrs.)

Diet: Probably a meat eater

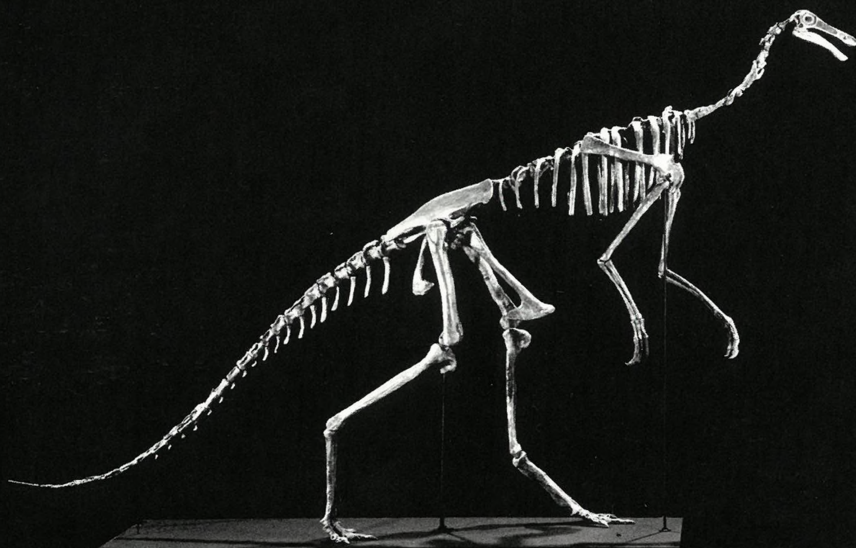
Size: 6 metres long

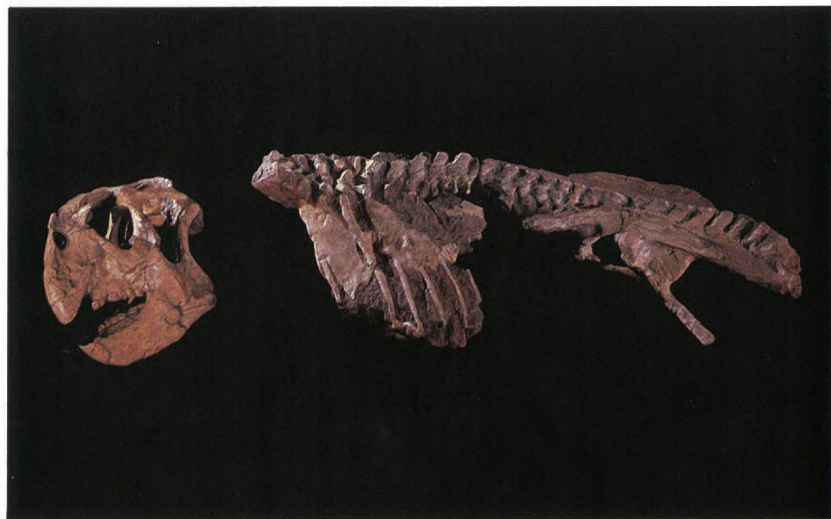
Exhibit: Near-complete skeleton (cast) found in the Gobi Desert, People's Republic of Mongolia, by a joint Polish-Mongolian expedition in 1968.

Other Finds: Three near-complete specimens from this region.

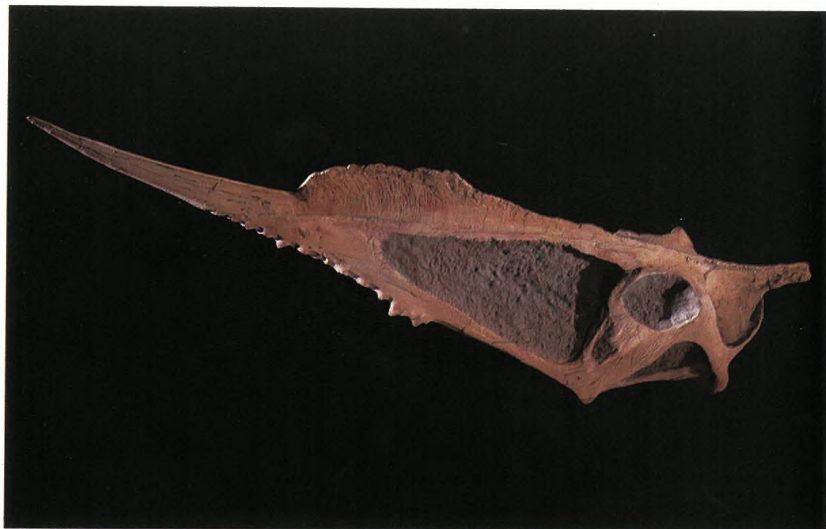
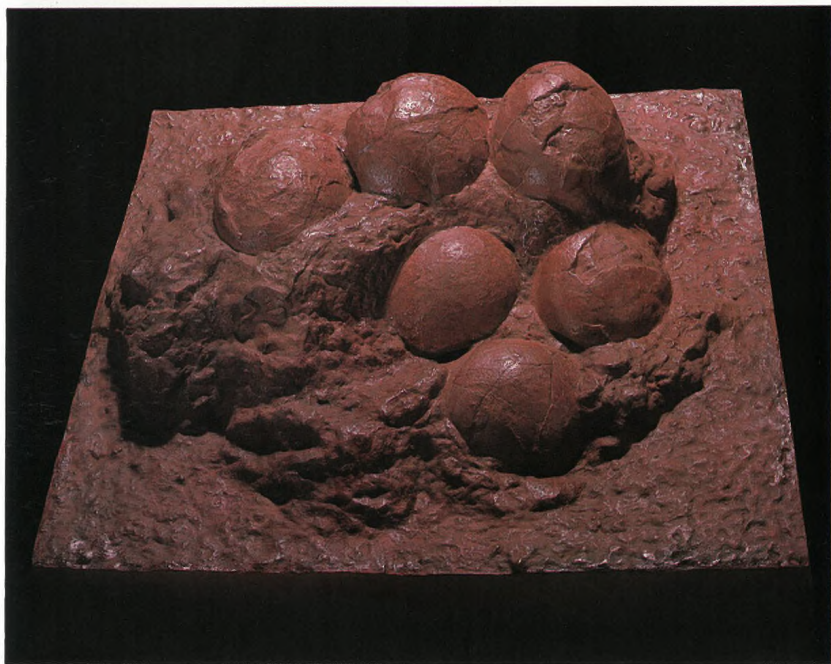
Comments: Its forelimbs end in long claws, which may have been used for raking or digging light material from the ground, rather than carrying food to the mouth.

It is a member of the group called coelurosaurs (see *Coelophysis*), which were slender, meat-eaters. It has long, striding hind-limbs, and as it walked, its tail probably swung free of the ground, perhaps aiding its balance.

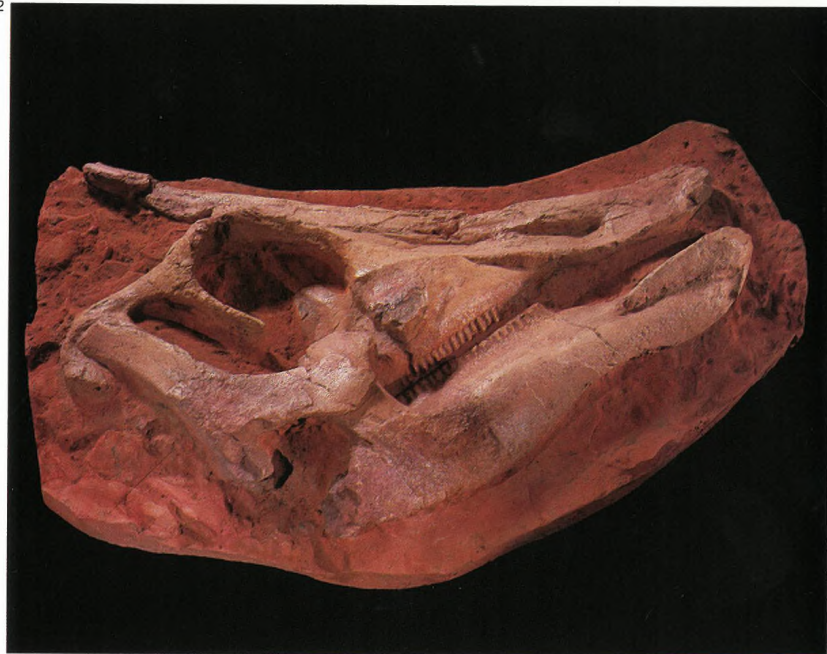




Psittacosaurus p.45
Dinosaur eggs



Dzungaripterus p.44
Saurolophus p.32



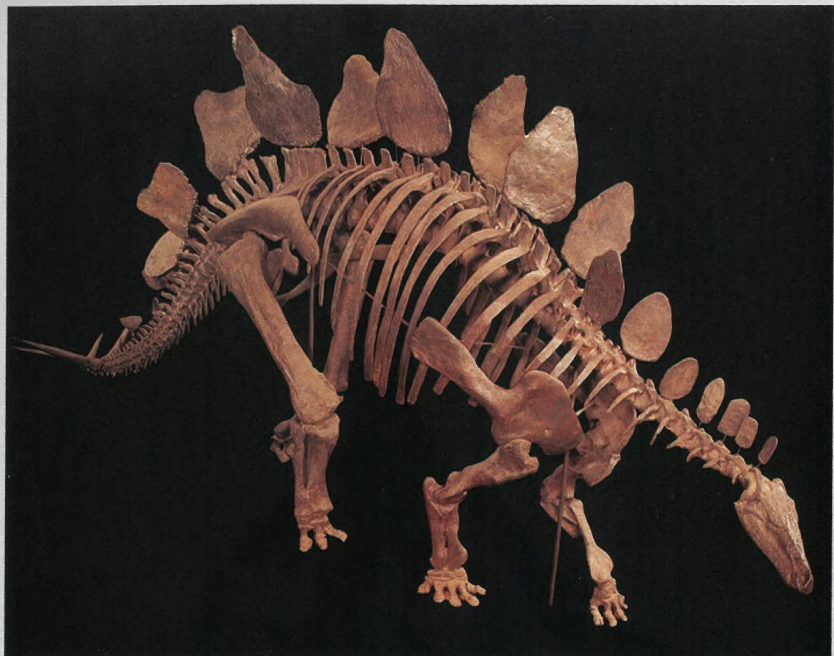


Tyrannosaurus p.42

Homalocephale p.33

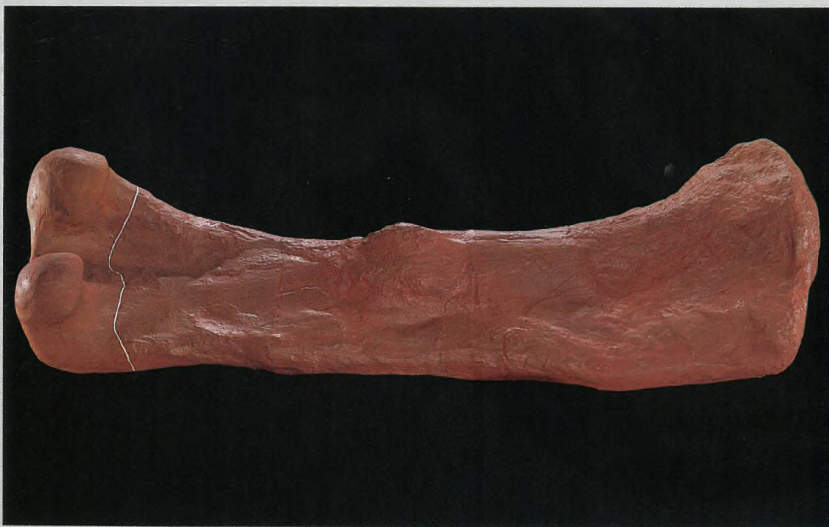


Tsintaosaurus p.20



Stegosaurus p.40

Rhoetosaurus p.35



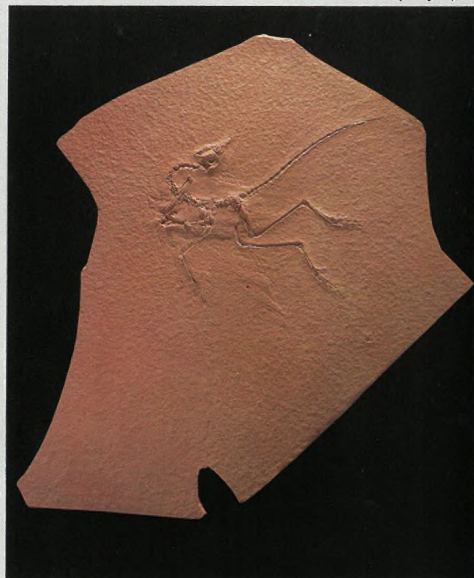
Deinococheirus p.43



Supersaurus p.34

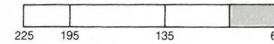
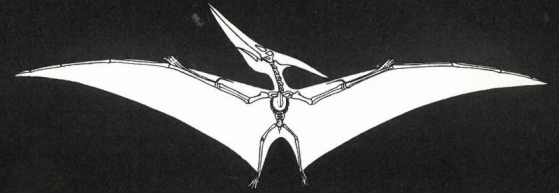


Archaeopteryx p.46



Footprint of a Carnosaur





Pteranodon (ter-an-o-don)

Meaning: 'Winged, Toothless'.

Age: Late Cretaceous (65-95 m.yrs.)

Diet: Fish-eater. Fragments of fish, fish-scales and crustaceans have been found in *Pteranodon* jaws and rib-cage.

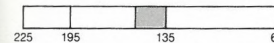
Size: Wingspan 7 metres (one of the largest known flying animals).

Exhibit: A restoration (model) based on many sources of information.

Other finds: Entire skeletons from the U.S.A. and U.S.S.R.

Comments: The breast-bone is relatively small, restricting the size of flight muscles, so presumably *Pteranodons* would have been able to glide and soar. The skull is large, but light-weight; the jaws were toothless, perhaps with a pelican-like pouch in the lower jaw. A distinctive feature is the crest extending from the top of the skull. Its function is uncertain; perhaps it was a counter-weight to the beak allowing for reduction of the size of neck muscles or perhaps a rudder for stability in flight.

Pteranodon was a more advanced pterosaur (flying reptile) than *Dzungaripterus* and *Rhamphorhynchus*. Having no tail and no teeth, it was adapted to a specialised mode of flying and feeding, perhaps similar to the living albatrosses.



Rhamphorhynchus (ram-fo-rink-us)

Meaning: 'Beak-nose'

Age: Late Jurassic (135-155 m.yrs.)

Diet: Probably a fish-eater

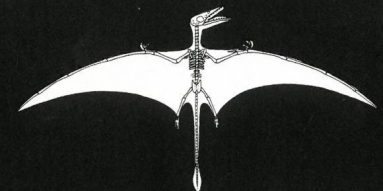
Size: 30 cm. long

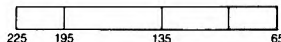
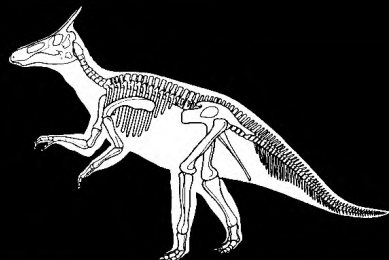
Exhibit: Restoration (model) of specimen found in U.S.S.R. in 1948 by A. N. Riabinin a Russian palaeontologist.

Other Finds: Near-complete skeletons found in lithographic limestone in southern Germany.

Comments: An early form of pterosaur (flying reptile). Its flying membrane was stretched between the 4th digit (finger) and its body, enabling similar flight to modern bats.

The skin flap at the end of its tail was held vertically and may have acted as a rudder. The head is large, with large eye sockets. Its numerous small teeth protrude forward and sideways, perhaps enabling it to snap up food and hold it while in flight.





Saurolophus (sawr-ah-lof-us)

Meaning: 'Reptile, crest', so named because of a feature on top of its skull.

Age: Late Cretaceous (65-95 m.yrs.)

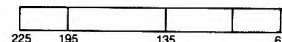
Diet: Plant-eater

Size: 10 metres long

Exhibit: Skull (cast) found in the Gobi Desert, People's Republic of Mongolia in 1969 by a joint Polish-Mongolian expedition.

Other Finds: Western Canada

Comments: It is a member of the group called the hadrosaurs, or 'duck-billed' dinosaurs. The skull has a solid spike-like crest, formed by an extension of the nasal bones rising high over the head. Its function is thought to have been for combat between males competing for females or to enable individuals of the same species to recognise each other. The top of the jaw is flattened at the end like a duck's bill, and some scientists believed it fed in shallow water as ducks do. There are no teeth at the front of the jaws, but at the back are masses of grinding teeth. In some specimens, signs of wear suggest that some hadrosaurs ate hard, coarse food. Fossilized stomach contents have been found, which include pine needles, twigs, seeds and fruit, suggesting that these dinosaurs sought food on dry land.



Homalocephale (hom-al-o-sef-al-ee)

Meaning: 'Even-head'

Age: Late Cretaceous (65-95 m.yrs.)

Diet: Plant-eater

Size: 1.5 metres high

Exhibit: Skull (cast) found in the Gobi Desert, People's Republic of Mongolia, by a joint Polish-Mongolian expedition of 1970.

Other Finds: A partial skeleton still in place.

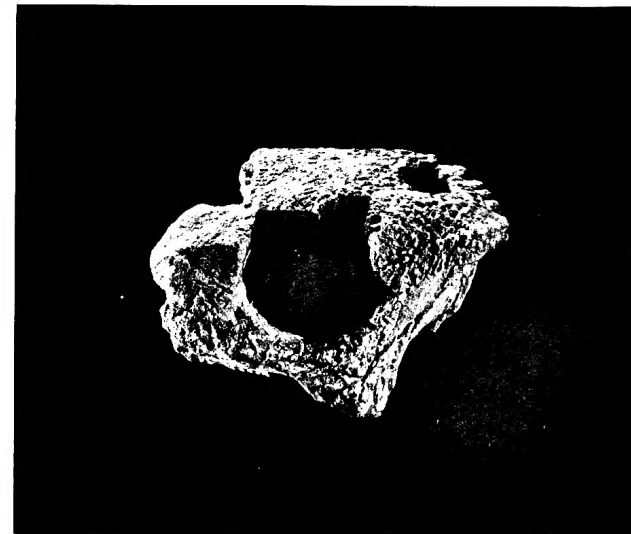
Comments: This is a member of a group called pachycephalosaurs ('thick-headed reptiles') so named because the roof of the skull, often dome-shaped, is up to 20 cm thick in some species.

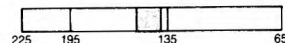
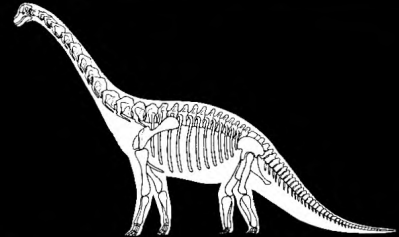
This feature may have been for defence, or for use in contests of sexual rivalry (the males head-butting each other, as male sheep and goats do).

Homalocephale is distinguished and named by the fact that its skull roof, while thick, is quite flat.

Large eye sockets suggest that its vision was well developed.

Pachycephalosaurs are thought to have been bipedal, plant-eating animals.





'Supersaurus' (super-sawr-us)

Meaning: 'Big Reptile' (not a scientific name, but in common use).

Age: Late Jurassic (140-155 m.yrs.)

Diet: Probably a plant-eater

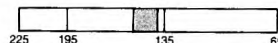
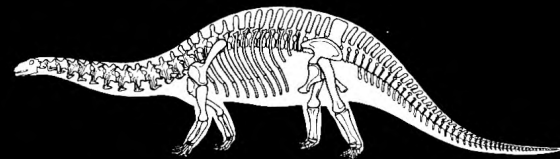
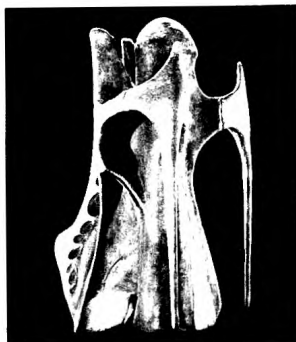
Size: 25-30 metres long (estimated)

Exhibit: Shoulder blade (scapula) and neck vertebra (casts) found in Colorado, U.S.A. in 1972 by Dr. J. Jensen of Brigham Young University, Provo, Utah.

Other Finds: Isolated ribs and vertebrae.

Comments: Presumed to be a sauropod, similar to one called *Brachiosaurus*, which had longer forelimbs than hind limbs.

Known only from a few fossil bones, '*Supersaurus*' is believed to be the largest dinosaur yet discovered possibly weighing more than 100 tonnes.



Rhoetosaurus (ray-tow-sawr-us)

Meaning: 'Rhoetos' was a giant in mythology

Age: Late Jurassic (140-155 m.yrs.)

Diet: Plant-eater

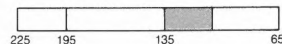
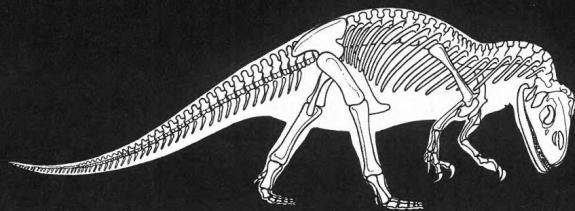
Size: 12.5 metres long (estimated)

Exhibit: Thigh-bone (cast) discovered near Roma, Queensland, in 1924 by A. J. Browne in association with Queensland Museum.

Other Finds: Series of vertebrae (bones of neck, back and tail), rib and pubic bones.

Comments: The fossil evidence shows it to have the form of a typical sauropod: it stood on four straight limbs in the posture of a modern elephant. Its neck was long. This and other features would have enabled *Rhoetosaurus* to feed either in swamps (on water plants) or on dry land, (perhaps browsing on trees), but scientists are not in total agreement about its feeding habits.





Allosaurus (al-lo-sawr-us)

Meaning: 'Other reptile', so named because its discoverer thought it so unlike his previous finds.

Age: Early Cretaceous (105-135 m.yrs.)

Diet: Meat-eater

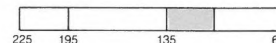
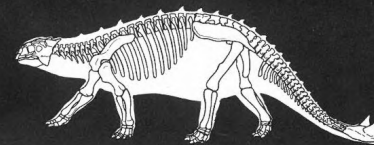
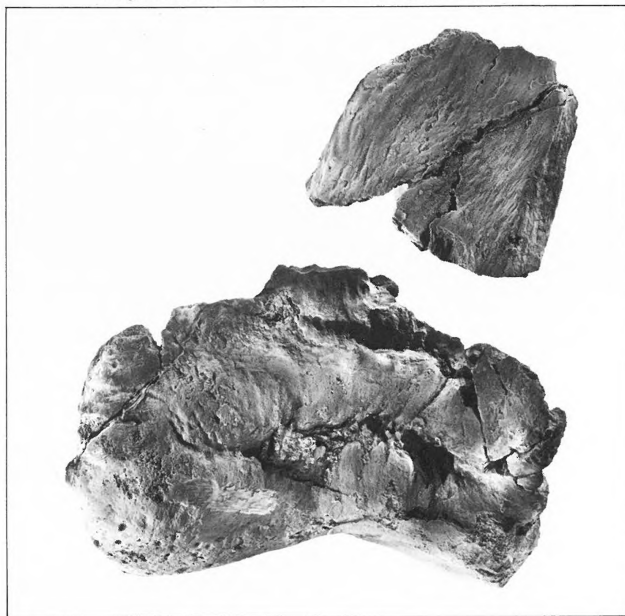
Size: Up to 11 metres long

Exhibit: A single, but highly distinctive foot-bone, found in the Strzelecki Ranges, Victoria, in 1978 by Mr. T. Flannery in association with the National Museum of Victoria.

Other Finds: The skeleton is well known from Late Jurassic finds in North America. *Allosaurus* has also been reported from Asia and Africa.

Comments: A bipedal dinosaur that probably was the major predator of its time. The structure of its skull and jaws enabled the mouth to open wide exposing dagger-like teeth. It had strong, heavy limbs and pelvis, and each of its short forelimbs ended in three large claws.

It was presumably capable of pursuing and attacking large prey, including *Stegosaurus* and sauropod dinosaurs like '*Supersaurus*'.



Ankylosauria (an-kyle-o-sawr-e-ya)

Meaning: 'Stiff Reptile'

Age: Early Cretaceous (105-135 m.yrs.)

Diet: Plant-eater

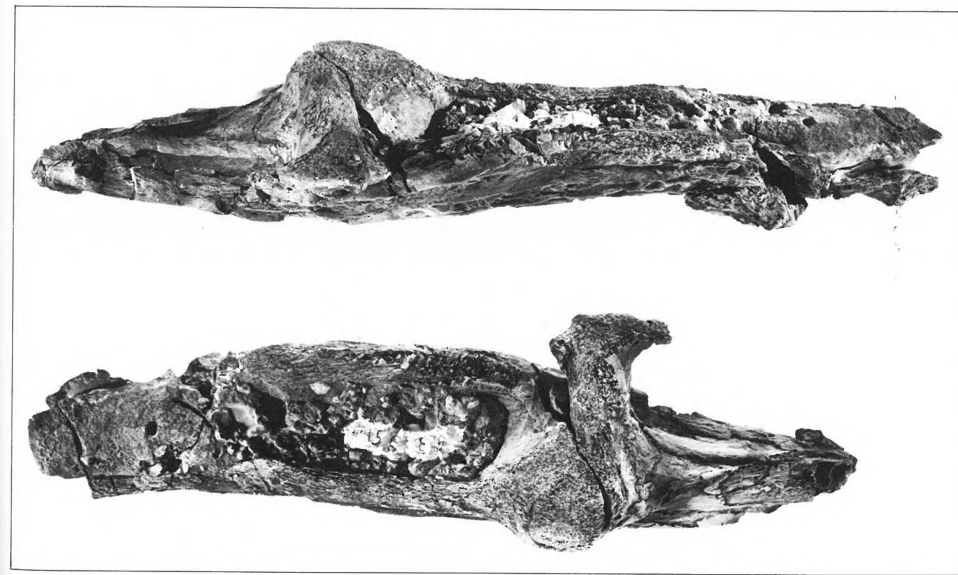
Size: 6 metres long

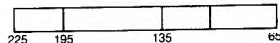
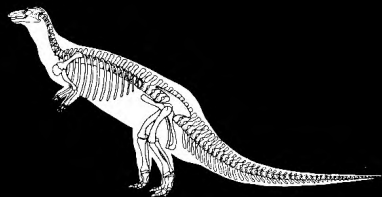
Exhibit: Fragment of lower jaw found at Cape Paterson, Victoria, in 1979 by Mr. T. Flannery and staff of the National Museum of Victoria.

Other Finds: Queensland, Asia, Europe, North America, South America.

Comments: The distinguishing feature of this group is their armoured skin. Unlike the shell of a turtle, it is flexible, covered with polygonal bony ossicles, with, here and there, larger bony bumps, flattened plates, scales or tall, cone-like spines. In life these were probably covered with a layer of horn. This armour probably evolved as a protection against powerful predators.

Ankylosaur teeth were small and weak. From this it may be inferred that they ate only soft plants. Ankylosaurs were quadrupeds (walking on four feet) with relatively short legs.





Camptosaurus (camp-toe-sawr-us)

Meaning: 'Bent reptile', so named because of extremely curved thigh-bones.

Age: Early Cretaceous (105-135 m.yrs.)

Diet: Plant-eater

Size: 2 to 5 metres long

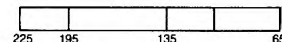
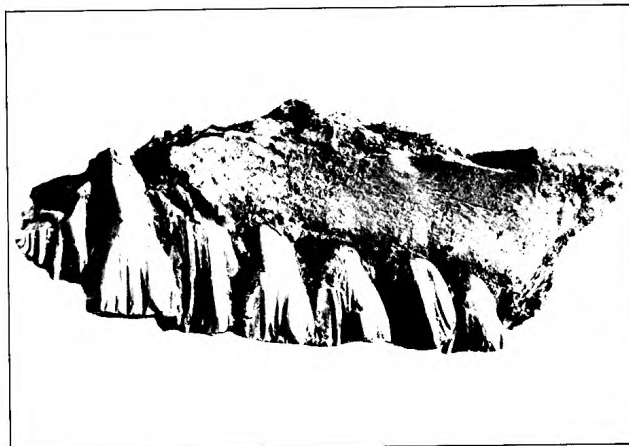
Exhibit: Skull fragment and thigh-bone found in the Otway and Strzelecki Ranges, Victoria in 1978 and 1979 by Messrs. T. Flannery and F. Bussat in association with staff of the National Museum of Victoria.

Other Finds: Skeletons from Late Jurassic of North America and Europe.

Comments: Its head was relatively small and the jaws have the appearance of a beak, which may have had a horny cover in life. Its cheek teeth are well developed, enabling it to chew plant material.

It walked on its hind limbs but the forelimbs were stout enough to support it while feeding. The claws of its five 'fingers' were flat and hoof-like, adapted more for walking than grasping.

The shorter tail was a less efficient counter balance so its posture when walking may have been more upright than that of other bipedal dinosaurs.



Hypsilophodon (hips-i-lof-o-don)

Meaning: 'High-crested teeth', a feature of this species.

Age: Early Cretaceous (105-135 m.yrs.)

Diet: Plant-eater

Size: 1-2 metres long

Exhibit: Thigh-bones found in the Strzelecki Ranges, Victoria, in 1979 by Mr. T. Flannery in association with staff of the National Museum of Victoria.

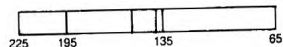
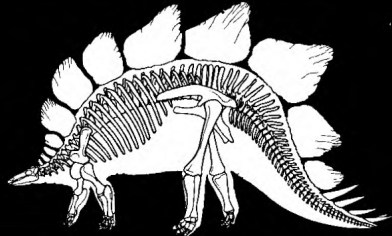
Other Finds: About 20 specimens, most being from the Isle of Wight, U.K., one from Utah, U.S.A. and a few from Lightning Ridge, N.S.W.

Comments: It stood on its hind legs, which are twice the length of the front legs. Its tail accounts for half the body length and was stiffened by bony tendons, presumably to hold it above the ground while running. Its hind feet were four-toed, while its 'hands' were five-fingered.

Its head is small and the jaws beak-like with only a few teeth in the upper jaw.

Early reconstructions by scientists showed *Hypsilophodon* as a tree-dweller balancing on and clinging to branches like modern tree-kangaroos. Not all scientists accept this interpretation today.





Stegosaurus (steg-o-sawr-us)

Meaning: 'Roofed reptile'; so named because of plates along its back.

Age: Late Jurassic (140-155 m.yrs.)

Diet: Plant-eater

Size: 9 metres long

Exhibit: Skeleton (cast) of specimen found in Wyoming, U.S.A. in 1901.

Other Finds: Skeletons have been found in North America, Europe, Asia and Africa.

Comments: Its skull is small, with a turtle-like beak and a few teeth, suggesting that its diet may have included soft plants such as fern fronds near swamps. Its size and bulk suggest that it stood on four legs. The spiked tail may have been used for defence. The plates on its back may be for protection, but it has recently been suggested that they may have been used to regulate the animal's body temperature, by presenting the large surfaces towards or away from sunlight as required.

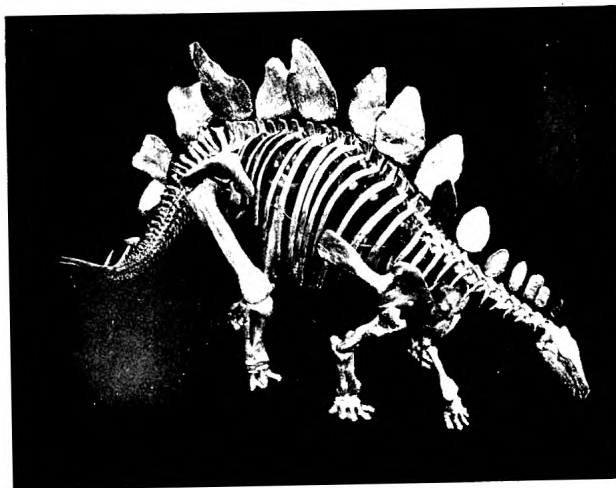
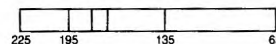


Photo Aus. Mus.



Coelophysis (see-low-fye-sis)

Meaning: 'Hollow-form'; a reference to its hollow, bird-like bones.

Age: Early Jurassic (170-180 m.yrs.)

Diet: Meat-eater

Size: 2.5 metres long

Exhibit: Skeleton-in-rock found at Ghost Ranch, New Mexico, U.S.A. in 1947 by staff of the American Museum of Natural History, New York.

Other Finds: Complete skeletons ranging from newly-hatched to adult specimens.

Comments: Its bones were all delicate with the long bones hollow like those of birds. The feet were bird-like, and the arms were well developed with 3-fingered hands. The head was long and tapered, and the teeth were thin and blade-like with finely serrated edges. Its entire skeleton seems to be designed to reduce weight and allow maximum activity.

It is presumed to have been a small, active predator, pursuing and grasping smaller animals. It is a member of the group called coelurosaurs, the most successful of all dinosaurs, in that they were among the first dinosaurs to appear and persisted until the time when all dinosaurs became extinct. A recently well-supported theory suggests that one branch of coelurosaurs developed feathers and eventually evolved into birds (see *Archaeopteryx*).

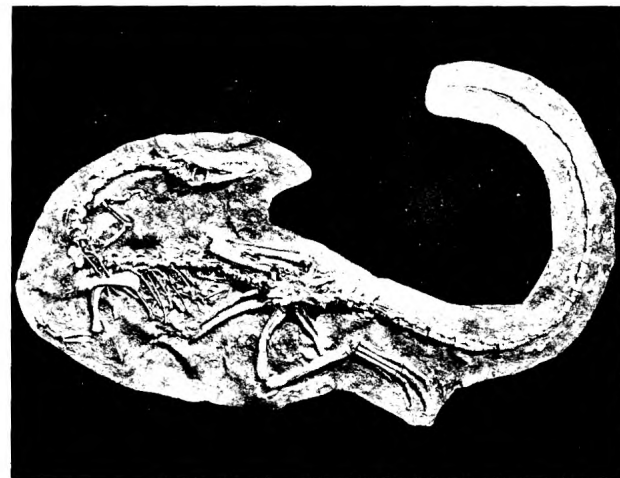
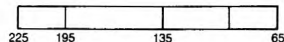
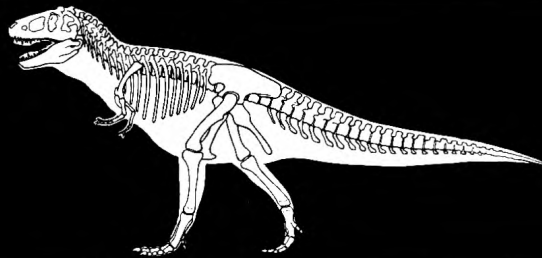


Photo Aus. Mus.



Tyrannosaurus rex (tye-ran-o-sawr-us)

Meaning: 'King of the tyrant lizard'

Age: Late Cretaceous (65-95 m.yrs.)

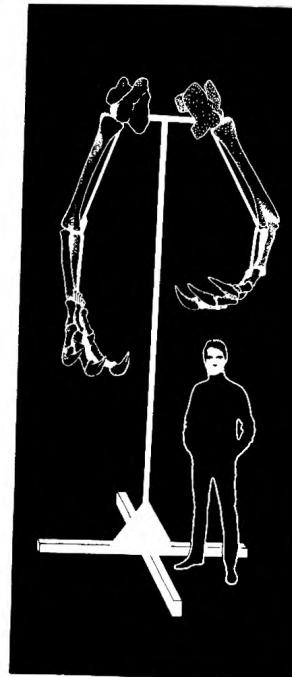
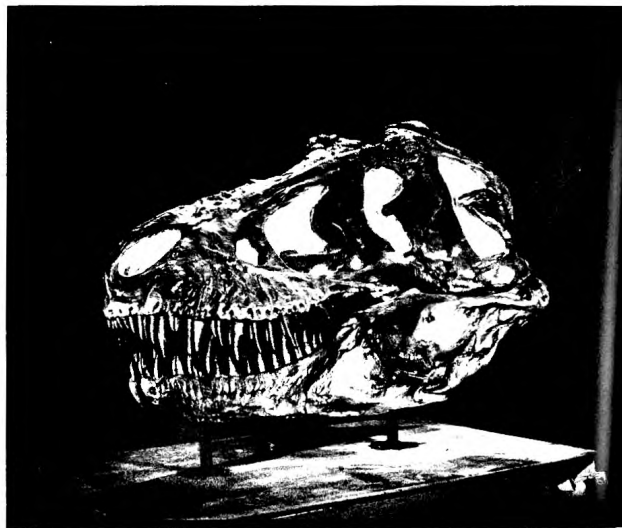
Diet: Meat-eater

Size: 10-14 metres long

Exhibit: Skull (cast) from specimen found in Montana, U.S.A., by Barnum Brown of the American Museum of Natural History, New York in 1902.

Other Finds: Complete skeletons in North America, Asia.

Comments: The structure of the skull and jaws allowed the mouth to open in a wide gape. The 15-20 cm dagger-like teeth have sharp serrated edges. It could have slashed at prey, or torn large chunks of meat from a carcass. Marks of these teeth have been found on fossil remains of other dinosaurs. Its likely prey was *Triceratops*, and sauropods such as *Alamosaurus*. Its small forelimbs, each ending in two claws, are thought to be too small to have had any grasping function. It walked on its hind legs its head low and tail outstretched for balance.



Deinocoelurus (dye-no-keer-rus)

Meaning: 'Terrible hand'

Age: Late Cretaceous (65-95 m.yrs.)

Diet: Probably a meat-eater

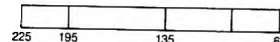
Size: Fore legs: 2.4 metres long

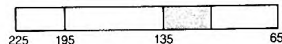
Exhibit: Left and right shoulder-blades and forelimbs (casts) found in the Gobi Desert, People's Republic of Mongolia, in 1965 by Professor Zofia Kielan-Jaworowska.

Other Finds: There are none to date.

Comments: These limbs represent the only fossil evidence of this dinosaur. It is presumed to be a predator similar to *Tyrannosaurus*, but its size is debatable. Its claws are the largest known of any dinosaur, being up to 30 cm long.

Unlike those of the forelimbs of *Tyrannosaurus*, however, they are powerful and may have been used for grasping and tearing at prey.





Dzungaripterus (zoon-gar-rip-ter-us)

Meaning: 'Dzungari' — region in China where first specimens were found.

Age: Early Cretaceous (105-135 m.yrs.)

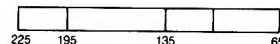
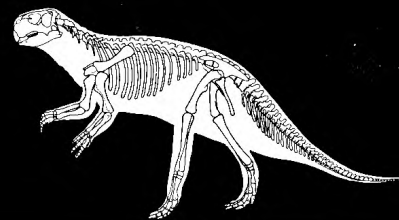
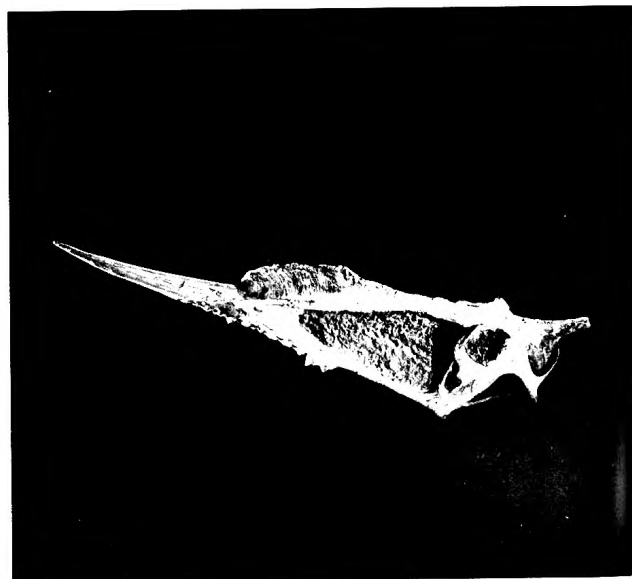
Diet: Probably a fish-eater

Size: Wingspan 3.5 metres

Exhibit: Skull (cast) found in the Dzungari Basin, Sinjiang, People's Republic of China by Professor C. C. Young.

Other Finds: Incomplete skeletons; imprints of skin, impressions of muscles and tendons adhering to bones; crinoids (sea lilies) among stomach remains.

Comments: *Dzungaripterus* is a pterosaur (flying reptile) that lies structurally intermediate between *Rhamphorhynchus* and *Pteranodon* in having teeth in the rear of the jaws, but not in the 'beak' part, which curves upwards like that of the avocet, a modern shorebird. Its diet evidently included fish and crinoids (sea lilies).



Psittacosaurus (sit-a-ko-sawr-us)

Meaning: 'Parrot-reptile', so named because of the shape of its skull.

Age: Early Cretaceous (105-135 m.yrs.)

Diet: Plant-eater

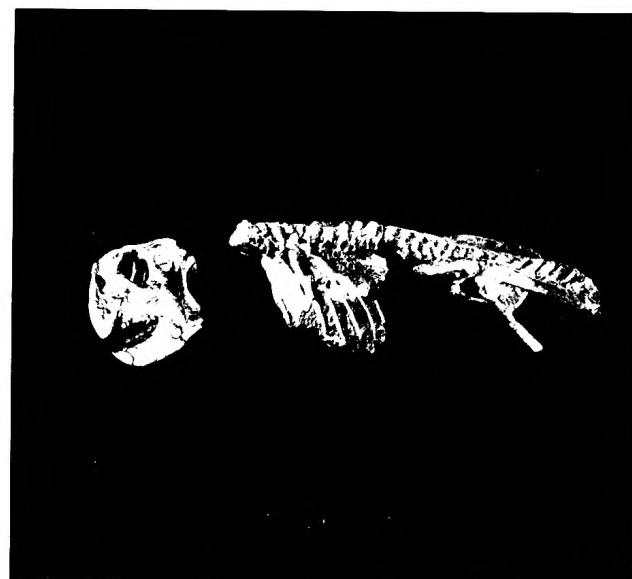
Size: 1.5-2.5 metres long

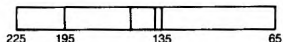
Exhibit: Skeleton in rock (cast) found in Shandong, People's Republic of China in 1960 by Professor C. C. Young of the Institute of Vertebrate Paleontology and Paleoanthropology.

Other Finds: Fossils including skulls, vertebrae and limb bones have been found in Mongolia and China.

Comments: It has long hind legs and presumably spent much of its time walking on two legs. The forelimbs are strong enough for walking on four legs, although the 'hands' may have retained some grasping function. It could have grazed on plants. The skull is narrow and rather deep with the snout ending in a sharp, parrot-like beak. Holes for the eyes and nostrils are placed high on the sides of the skull.

The small, flat crest at the back of the head suggests that it may be related to the horned dinosaurs or ceratopsians.





Archaeopteryx (ar-kee-op-ter-ix)

Meaning: 'Ancient Wing'

Age: Late Jurassic (140-155 m.yrs.)

Diet: Probably an eater of insects, fish or other small vertebrates.

Size: 21.5 cm long (about the size of a raven) Eichstätt *Archaeopteryx*
34 cm long Berlin *Archaeopteryx*.

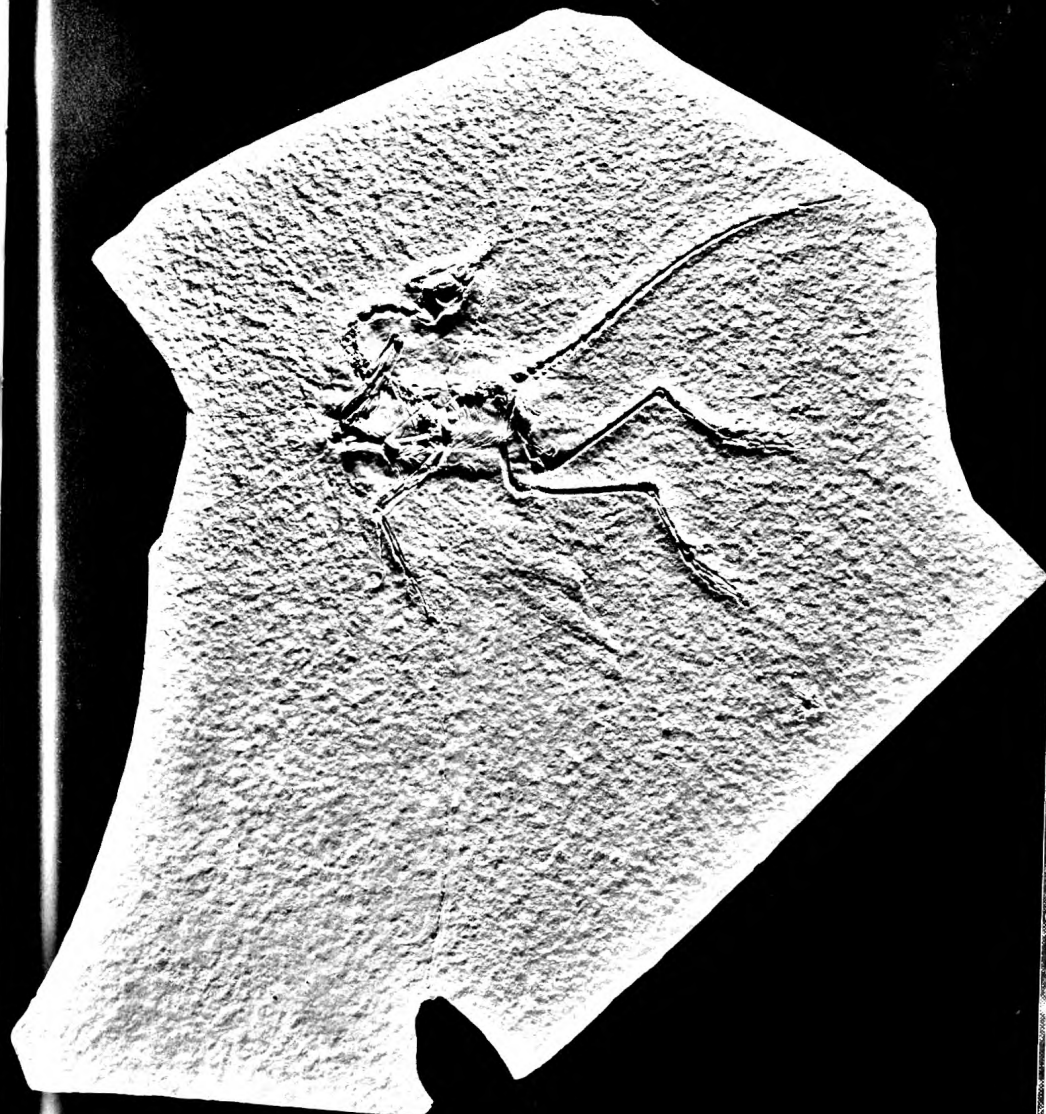
Exhibit: Skeletons in rock (cast) found in Bavaria, southern Germany. The specimen housed in a Berlin museum was found in 1877. The specimen housed in a museum in Eichstätt was found in 1951.

Other Finds: From the lithographic limestone quarries in Bavaria, southern Germany 5 specimens have been recovered. Some specimens have impressions of feathers.

Comments: This "reptile" had, like modern birds, feathers, a wishbone, wings and a small breastbone. It is assumed to have been capable of at least poor, gliding flight. It also has, like reptiles, scales on its legs and head, a long bony tail, teeth, three-clawed forelimbs, and a small, reptile-like brain (but large within reptiles). This lends weight to the theory that modern birds are the descendants of dinosaurs.

It is assumed to be closely related to coelurosaurian dinosaurs (See *Coelophysis*).

The Eichstätt specimen was originally thought to be a small coelurosaur, until feather impressions were noticed some 20 years after its discovery.



Moulding and Casting of Dinosaur Bones

One of the main activities, undertaken while the two Chinese dinosaur specimens are at the National Museum of Victoria, is taking moulds of all the bones and preparing casts from these moulds. This process has at least a dozen separate steps and takes several days for each bone. Each of the two skeletons has about two hundred or more bones and for many of these moulds consisting of more than one piece will have to be taken because of their complex shape. This is especially true for such parts as the skulls and vertebrae.

For this major undertaking one public gallery at the National Museum has been closed and converted into a workshop for about a year. The moulding and casting is under the direction of the Museum's Senior Preparator, Mr. M. Traynor. It is being carried out with the assistance of several scientists and technicians from the Institute of Vertebrate Paleontology and Paleoanthropology, Academia Sinica, China and the Museum's preparatory staff.

The moulds are made from a newly developed medium, Silicone Rubber Q33321 generously supplied by Dow Corning (Australia). The casts are then manufactured using polyurethane foam.

The dozen steps used in making the mould and the cast are listed below.

1 Planning the Mould

Before making a mould it is important to study closely each of the fossil bones. The first step is to make a plasticine wall and base around the bone to contain the flow of the liquid silicone rubber that will form the mould.

2 Building the Dividing Wall

Plasticine is placed along the mid-line of the bone to form a wall which will hold the liquid silicone rubber until it sets.



3 Application of Silicone Rubber

Silicone rubber is used for making the mould because it is strong, elastic and quick to set and does not adhere to bone. Before the silicone rubber is poured it is mixed with a curing or setting agent. The liquid rubber is then carefully poured over the bone. A hot-air blower is used to partly set the rubber and prevent it flowing too freely.



4 Application of Gauze

Before the rubber has completely set it is reinforced by placing small pieces of gauze over its surface.

5 Final Coat of Silicone Rubber

A final coat of silicone rubber is poured over the gauze. Small, prefabricated silicone keys are then pressed into the walls of the mould. Set in the rubber these keys will serve to locate the sections of the mould and restrict its movement during casting.



6 Application of Plaster

A hard plaster jacket is now fabricated over the rubber mould. A thin coating of plaster is first applied then reinforced with sisal fibre and metal supports. Once the plaster has set hard the mould and bone are turned and work begins on the unmoulded side of the bone.

7 Use of Separator

Before moulding the second side, petroleum jelly is applied to any exposed surfaces of silicone rubber. This will ensure the clean separation of the two sides of the mould.



8 Completing the Mould

Using silicone rubber, gauze, and a plaster jacket as previously described in steps 1-7, the bone is entirely encased, and the mould completed.

9 Removing the Fossil

The sections of the mould are now prised apart. The bone is removed and cleaned.

10 Preparing the Mould for Casting

The mould sections are now placed together and secured firmly by wire or clamps and the mould is ready to make a cast of the bone.



11 Casting

The material used in casting is a polyurethane foam. The two ingredients of this casting medium, one of which is a blowing or foaming agent, are rapidly mixed then poured into the mould through a special pouring hole. Within two minutes foaming takes place in the mould. A plug or stopper is applied to the pouring hole to prevent the foam from escaping. The pressure created ensures that the expanding foam takes in all the detail of the mould.



12 Removal of Cast

After forty minutes the foam has hardened. The cast can now be removed from the mould trimmed and coloured. To even the most practised eye the cast will be indistinguishable from true bone.

Suggestions for Further Reading

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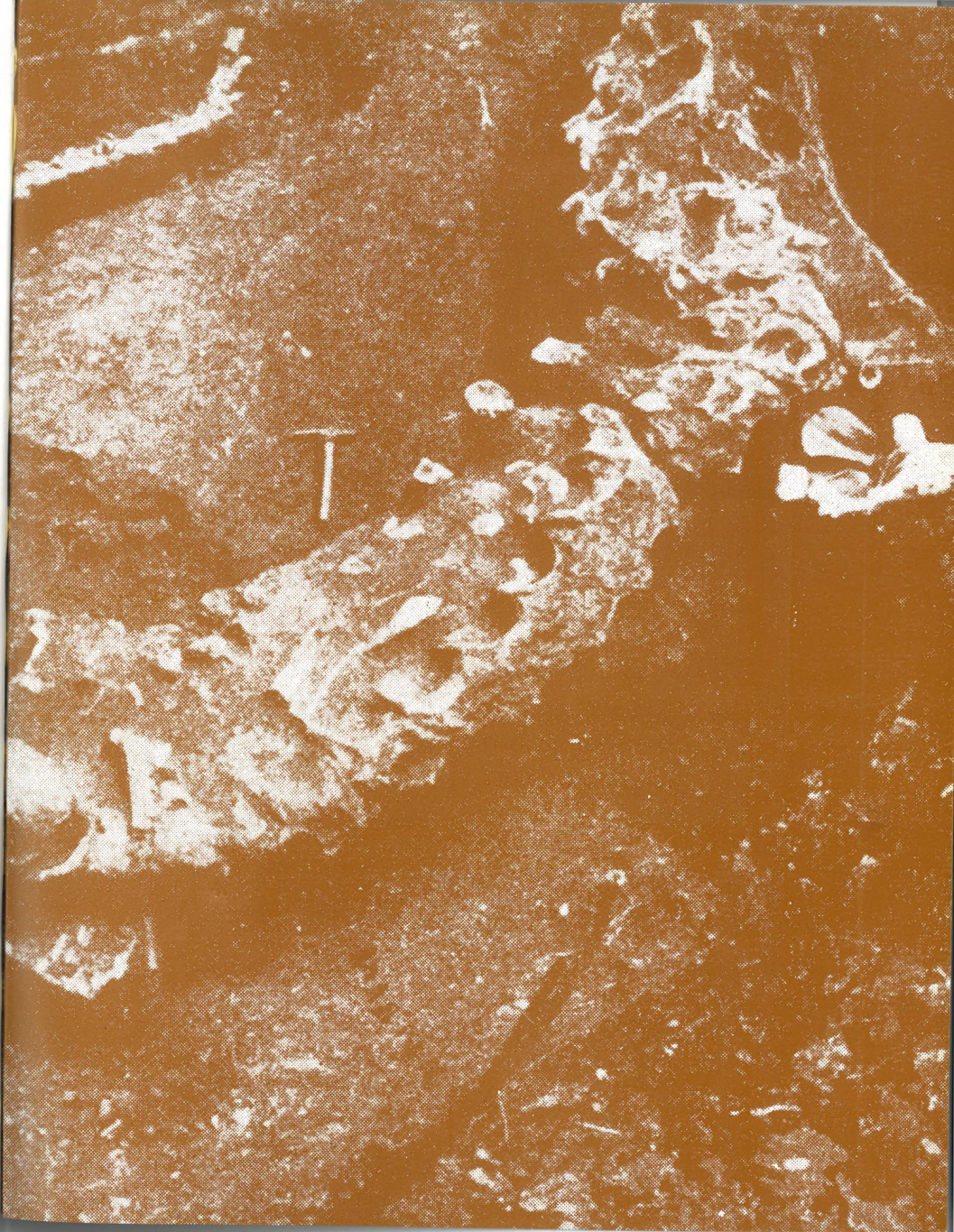
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